

English Version

**A framework for digital preservation of cinematographic
works - The Cinema Preservation Package**

Un cadre pour la conservation numérique des oeuvres
cinématographiques - Le paquetage de conservation
cinéma

Ein Rahmenwerk für die digitale Erhaltung von
kinematografischen Werken - Das Cinema
Erhaltungspaket

This draft European Standard is submitted to CEN members for formal vote. It has been drawn up by the Technical Committee CEN/TC 457.

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European foreword

This document (FprEN 17650:2022) has been prepared by Technical Committee CEN/TC 457 “Digital preservation of cinematographic works”, the secretariat of which is held by DIN.

This document is currently submitted to the Formal Vote.

The European Project targets the publication of a Technical Report complementing this document.

Introduction

This document is part of a series of standards and technical recommendations for the digital preservation of cinematographic works. It gives European film archives and producers a guideline for storing and managing films in the digital age. The document references many existing formats and elements and serves as a super-format which includes other existing sub-formats like DCPs or IMPs for movies, XML-files for packing-lists or subtitles, AV files and metadata files. In addition, methods are defined to ensure data integrity and quality by calculating hash and fingerprinting-values.

This document reuses describing tools and archiving formats from the cinema archiving community as much as possible.

1 Scope

This document specifies the Cinema Preservation Package (CPP) to facilitate the digital preservation of cinematographic works. It specifies methods to describe the relationship of components of a cinematographic work and delivers a syntax to describe the package content. The document itself specifies the structure of the package and the constraints that are necessary to enable compliance and interoperability.

Versions of the content using different encoding formats can be preserved in a layered structure where the lowest level is describing the physical file. The files can carry data representing moving images, sound, metadata or ancillary information like quality control (QC) reports or film posters.

The Cinema Preservation Package also contains hash values on different levels to ensure data integrity and version control. The syntax for this description and the methods for the hash value generation are specified in this document. Various types of packages are described as reference for practical implementations.

The Cinema Preservation Package is designed to serve as a Submission Information Package (SIP) in an OAIS compliant preservation system, and it may be used as a self-contained exchange format between media archives (as Distribution Information Package DIP). Based on the requirements of an archive, it can also be used as complete Archival Information Package (AIP) if the constraints set out in this standard for such use are adhered to.

A CPP does not necessarily contain a complete cinematographic work, it can also be used for the exchange of parts of a work.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 15744, *Film identification — Minimum set of metadata for cinematographic works*

EBU Tech 3264-E, *Specification of the EBU Subtitling data exchange format*

EBU Tech 3285, *Specification of the Broadcast Wave Format (BWF)*, Geneva, May 2011

EBU Tech 3293, *EBU Core Metadata Set (EBUCore), Specification v. 1.10*, Geneva, April 2020

EBU Tech 3350, *EBU-TT, part 1 subtitling format definition*, Geneva, May 2017

ETSI TS 102 366, *Digital Audio Compression (AC-3, Enhanced AC-3) Standard*, September 2017

IANA. *Media Types* [online], Available from <https://www.iana.org/assignments/media-types/media-types.xhtml>

IEEE/Open Group 1003.1-2017, *IEEE Standard for Information Technology — Portable Operating System Interface (POSIX(TM)) Base Specifications, Issue 7*

IETF RFC 9043, *FFV1 Video Coding Format Versions 0, 1, and 3*

ISO 639-1, *Codes for the representation of names of languages — Part 1: Alpha-2 code*

ISO 639-2, *Codes for the representation of names of languages — Part 2: Alpha-3 code*

ISO 639-3:2007, *Codes for the representation of names of languages — Part 3: Alpha-3 code for comprehensive coverage of languages*

ISO 8601-1, *Date and time — Representations for information interchange — Part 1: Basic rules*

ISO 8601-2, *Date and time — Representations for information interchange — Part 2: Extensions*

ISO 12234-2, *Electronic still-picture imaging — Removable memory — Part 2: TIFF/EP image data format*

ISO 19005-1, *Document management — Electronic document file format for long-term preservation — Part 1: Use of PDF 1.4 (PDF/A-1)*

ISO 19005-2, *Document management — Electronic document file format for long-term preservation — Part 2: Use of ISO 32000-1 (PDF/A-2)*

ISO 19005-4, *Document management — Electronic document file format for long-term preservation — Part 4: Use of ISO 32000-2 (PDF/A-4)*

ISO 26429 (all parts), *Digital cinema (D-cinema) packaging*

ISO/IEC 9834-8, *Information technology — Procedures for the operation of object identifier registration authorities — Part 8: Generation of universally unique identifiers (UUIDs) and their use in object identifiers*

ISO/IEC 10646, *Information technology — Universal Coded Character Set (UCS)*

ISO/IEC 10918 (series), *Information technology — Digital compression and coding of continuous-tone still images*

ISO/IEC 13818-2, *Information technology — Generic coding of moving pictures and associated audio information — Part 2: Video*

ISO/IEC 13818-3, *Information technology — Generic coding of moving pictures and associated audio information — Part 3: Audio*

ISO/IEC 14496-3, *Information technology — Coding of audio-visual objects — Part 3: Audio*

ISO/IEC 14496-10, *Information technology — Coding of audio-visual objects — Part 10: Advanced Video Coding*

ISO/IEC 14496-12, *Information technology — Coding of audio-visual objects — Part 12: ISO base media file format*

ISO/IEC 14496-14, *Information technology — Coding of audio-visual objects — Part 14: MP4 file format*

ISO/IEC 15444-1, *Information technology — JPEG 2000 image coding system — Part 1: Core coding system*

ISO/IEC 15948, *Information technology — Computer graphics and image processing — Portable Network Graphics (PNG): Functional specification*

ISO/IEC 23091-3:2018, *Information technology — Coding-independent code points — Part 3: Audio*

ITU-T H.273:2016, *Coding-independent code points for video signal type identification*

ITU-R BS.1352-3, *File format for the exchange of audio programme materials with metadata on information technology media*

SMPTE RDD 36, *SMPTE Registered Disclosure Doc — Apple ProRes Bitstream Syntax and Decoding*

SMPTE RDD 45, *SMPTE Registered Disclosure Doc — Interoperable Master Format — Application ProRes*

SMPTE ST 12-1, *SMPTE Standard — For Television — Time and Control Code*

SMPTE ST 268, *SMPTE Standard — File Format for Digital Moving-Picture Exchange (DPX)*

SMPTE ST 428-7, *SMPTE Standard — Digital Cinema Distribution Master — Subtitle*

SMPTEST 429 (all parts), *SMPTE Standard - D-Cinema Packaging*

SMPTE ST 2019-1, *SMPTE Standard — VC-3 Picture Compression and Data Stream Format*

SMPTE ST 2065-4, *SMPTE Standard — ACES Image Container File Layout*

SMPTE ST 2067-3:2020, *SMPTE Standard — Interoperable Master Format — Composition Playlist*

SMPTE ST 2067-21, *SMPTE Standard — Interoperable Master Format — Application #2E*

SMPTE ST 2067-40, *SMPTE Standard — Interoperable Master Format — Application #4 Cinema Mezzanine*

SMPTE ST 2067-50, *SMPTE Standard — Interoperable Master Format — Application #5 ACES*

W3C - *Extensible Markup Language (XML) 1.1 (Second Edition)*

W3C - *W3C XML Schema Definition Language (XSD) 1.1 Part 1: Structures*

W3C - *W3C XML Schema Definition Language (XSD) 1.1 Part 2: Datatypes*

W3C - *XML Path Language (XPath) 3.0*

W3C - *TTML Profiles for Internet Media Subtitles and Captions 1.1*

Matroska MKV format specification. Available from: <https://www.matroska.org>

METS Metadata Encoding & Transmission Standard. Version 1.12.1. [online]. Library of Congress, 2019. Available from: <https://www.loc.gov/standards/mets/>

PREMIS DATA DICTIONARY FOR PRESERVATION METADATA. Version 3.0. [online]. Library of Congress, 2015. Available from: <https://www.loc.gov/standards/premis/>

SCHEMA P.R.E.M.I.S. 3.0 [online]. Library of Congress, 2015. Available from: <https://www.loc.gov/standards/premis/v3/premis-v3-0.xsd>

Xiph FLAC format specification. Available from: <https://xiph.org/flac/format.html>

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at <https://www.iso.org/obp>

— IEC Electropedia: available at <https://www.electropedia.org/>

3.1

audio channel

sequence of data representing a single audio signal

3.2

audio track

part of a sound file containing one or more audio channels or multiple single audio channels files forming one sound sequence

3.3

cinema preservation package

package containing digital data of a cinematographic work

3.4

container

file containing one or more data objects wrapped in a specific way

3.5

package

set of files or folders belonging together

3.6

provenance

metadata related to the origin, modification and the ownership history of a work

3.7

soundfield

three-dimensional pressure distribution produced by transmitted sound energy

NOTE A soundfield can be represented as a discrete set of signals, monophonic or multichannel, or other representation.

3.8

sound sequence

continuous group of sound samples related to a specific period of time

3.9

submission information package

package sent from the data producer to the archive

NOTE The submission information package is specified in the ISO 14721:2012 OAIS Model.

3.10

subpackage

set of data in a Cinema Preservation Package containing part of a cinematographic work with data and supplemental data separately stored in a folder

3.11

subpicture

picture to be overlayed on the main picture

4 Abbreviations

ACES	Academy Color Encoding System
ASCII	American Standard Code for Information Interchange
BMFF	Base Media File format
CPP	Cinema Preservation Package
DCP	Digital Cinema Package
DPX	Digital Picture Exchange
FLAC	Free Lossless Audio Codec
IMF	Interoperable Master Format
JPEG	Joint Photographic Experts Group
LoC	Library of Congress, see https://www.loc.gov/
MPEG	Moving Pictures Experts Group
OAIS	Open Archive Information System
PCM	Pulse code modulation
PDF	Portable Document Format
PNG	Portable Network Graphics
SIP	Submission Information Package
TIFF	Tagged Image File Format
UUID	Universal Unique Identifier
XML	Extensible Markup Language

5 Syntax conventions used in this document

5.1 General

For the file and folder name descriptions the following conventions are used.

5.2 Expressions used to denote file or folder names

5.2.1 Composition of name with parts

A name may be composed of one or several parts.

5.2.2 Literal parts

A literal part appearing between quotes shall appear as is.

EXAMPLE “LiteralPart” designates a part which is exactly “LiteralPart”.

5.2.3 Optional part

A part appearing between brackets shall appear zero or one time.

EXAMPLE [“OptionalPart”]“Radical” designates a name which is either “OptionalPartRadical” or “Radical”.

5.2.4 Alternative parts

Exactly one of the items in parentheses, separated by a vertical line '|', shall appear in the result.

EXAMPLE ("Prefix1"|"Prefix2"|"Prefix3")"Radical" designates a name which is either "Prefix1Radical", "Prefix2Radical" or "Prefix3Radical".

5.2.5 Explicitly defined part

<any text here> is a placeholder text and shall be replaced as specified by the prose.

EXAMPLE "Name_"<custom identifier> designates a name which could be "Name_0123456789", supposing "0123456789" is defined as a valid custom identifier.

5.3 Expressions used to denote file or folder multiplicity

5.3.1 General

Some files or folders may appear several times in the parent folder. A trailing modifier indicates the allowed multiplicity.

5.3.2 One occurrence or more

A file or folder name with a trailing '+' shall appear one or more times in the parent folder.

EXAMPLE "File_"<number>+ designates a set of multiple files such as "File_0000", or "File_0000" and "File_0001" and "File_0002", etc., assuming the prose describes such a 4-digit numbering.

5.3.3 Zero occurrence or more

A file or folder name with a trailing '*' shall appear zero or more times in the parent folder.

EXAMPLE "File_"<number>* designates a set of multiple files such as no file, "File_0000", or "File_0000" and "File_0001" and "File_0002", etc., assuming the prose describes such a 4-digit numbering.

5.3.4 Optional file or folder

A file or folder name with a trailing '?' shall appear zero or one time in the parent folder.

EXAMPLE "OptionalFile"? designates no file, or exactly one file with name "OptionalFile".

5.4 Typographic conventions

5.4.1 Monospaced fonts

In the document, monospaced fonts are used for literal expression of textual content of a file, for instance as example quotation.

EXAMPLE <exampleXmlElement>example text</exampleXmlElement>.

5.4.2 Italic

In the document, italic text is expected to be substituted as specified in the prose.

EXAMPLE *relative path to file* shall be the path from the root folder to the Descriptive Metadata file including the filename.

5.5 Conventions used to denote XML content

5.5.1 General consideration

Constraints on XML files are specified either as an XML Schema to apply, or as listed constraints for each element of a list of specific XML Nodes.

5.5.2 Hierarchy

To denote the hierarchical position of a node the XPath syntax is used.

EXAMPLE “/rootElement/subElement/subSubElement” denotes an element with name “<subSubElement>” which is a child of a “<subElement>” element, which itself is a child of the “<rootElement>” root element.

NOTE When an element in the path occurs more than one time, the XPath will be completed with discriminatory attributes if necessary. If the description applies to multiple elements, no discriminatory attribute will be added.

5.5.3 Namespace

XML elements may use a prefix to denote a namespace. The prefix indicated in the prose is only used for convenience, and does not denote a namespace. The prefix in the file may be different and this is suitable as long as the relation with the namespace is correctly indicated as a namespace attribute from a parent element.

EXAMPLE “/mainNs:rootElement/mainNs:subElement” the “<rootElement>” and the “<subElement>” belong to the namespace denoted by “mainNs” in the prose.

5.5.4 Subelements

The following are used:

- @attributeName: denotes the name of an attribute of the element
- #children: denotes the child elements
- #text: denotes the child text of the element

EXAMPLE

```
<mainNs:rootElement attribute1="example 1">
```

```
    <subElement1/>
```

```
    <subElement2>example 2</subElement2>
```

```
</mainNs:rootElement>
```

“/mainNs:rootElement/#children” denotes “subElement1” or “subElement2”

“/mainNs:rootElement/@attribute1” denotes “attribute1” with value “example 1”

“/mainNs:rootElement/subElement2/#text” denotes “example 2”

5.5.5 Property

If the XPath includes a part inside brackets, this is a constraint to properly address the right element.

EXAMPLE “/mainNs:rootElement/subElement[@position='first']” denotes the <subElement> element which is a child of the root element <rootElement> and has an attribute “position” with value “first”.

5.5.6 Conventions used in XML constraints list tables

Prior to each table, the current XML element is reminded using the XPath syntax.

In the following sections, the presence of elements is specified using tables. In these tables the “Element Arity” column indicates the multiplicity allowed for an element:

- 1: the element is mandatory, exactly one element shall be present;
- 0..1: the element is optional, it shall appear zero or one time;
- 1..n: multiple mandatory. The element shall appear at least one time and may be repeated;
- 0..n: multiple optional. The element might not be present, or might be present one or multiple times.

For each element subelements are indicated, the “Subelement mandatory” column indicates if this subelement is mandatory. An ‘x’ in the corresponding cell indicates that the subelement shall be present, an ‘o’ in the same column indicates the subelement may be present. An attribute starts with the character ‘@’. #text denotes the child text, if mandatory, the element cannot be empty. #children indicates xml child elements.

Aiming to reduce the number of tables and sections in this document, when the semantic allows to do so consistently, subelements are sometimes documented in the same table when only few elements are described.

EXAMPLE ‘title/dc:title’ indicates that the subelement <title> contains itself a sub-subelement <dc:title>, meaning that the whole XML segment has to be added to the current element described.

```
<title typeLabel="titleType">
  <dc:title lang="fr"> title name</dc:title>
</title>
```

6 Core structure

6.1 General description (informative)

A Cinema Preservation Package is a set of files related to a cinematographic work.

The package is composed of subpackages, each of them including an atomic sequence of media, and replicating a similar structure with metadata and a Packing List file including checksums. Package related data and subpackages are located in subfolders of the root folder.

The Preservation Packing List file, at the root, lists the subfolders and checksums of the Subpackage Packing List files.

6.2 General constraints

The Cinema Preservation Package shall be a set of files organized as described in this document.

The stored items shall be related to the same work and variant.

Metadata files shall be located at the root of the file tree and in designated subdirectories.

6.3 Root Folder structure

6.3.1 Synthetic view (informative)

The root folder contains the following files and subfolders:

- *<Preservation Package Instance name>*;
- “preservationPackingList.xml”;
- “playlists”?;
- (“audiovisual”|“componentized”|“image”|“sound”|“timedText”|“extra”)“Package_”*<uuid>*+;
- “metadata”;
- “ancillaryData”?;
- “checkerReports”?.

6.3.2 Root folder name

The root folder name designated as *<Preservation Package Instance name>* should follow the pattern:
<titleName>[“_”*<versionName>*][“_”*<work identifying system name>*“_”*<work identifier>*]

Where:

- *<titleName>* should be the title of the work, using alphanumerical ASCII characters, with ‘-’ as word separator;
- *<versionName>* if present should be the designation of the version of work, typically a variant, using alphanumerical ASCII characters, with ‘-’ as word separator;
- *<work identifying system name>* should be a prefix of four alphabetical upper-case characters. If the identifying system is an archive private system, then the name should be ‘PRIV’;
- *<work identifier>* should be the identification of the work in the system denoted by the preceding prefix, using only alphanumerical ASCII characters and ‘-’.

NOTE The preservation facility can use the identification of the work from their local catalogues as work identifier.

EXAMPLE “Goal-The-dreams-begins_feature_ISAN_0000-0001-8947-0000-8-0000-0000-D”.

6.3.3 Root folder content

6.3.3.1 Contained file

The root folder shall contain exactly one file.

The “preservationPackingList.xml” document shall be present and shall conform to provisions specified in 8.3.

6.3.3.2 Subpackage folders

One or more subpackage folders shall be present as children of the root folder. Each folder is a Subpackage and shall contain either an Image Package, a Sound Package, a Timed Text Package, an Audiovisual Package, a Componentized Package or an Extra Package, together with its related content.

Each subpackage folder shall conform to structure provisions as specified in 6.4 and conform to specific provisions:

- each Image Package folder shall conform to other provisions as specified in 9.1;
- each Sound Package folder shall conform to other provisions as specified in 9.2;
- each Timed Text Package folder shall conform to other provisions as specified in 9.3;
- each Audiovisual Package folder shall conform to other provisions as specified in 9.4;
- each Componentized Package folder shall conform to other provisions as specified in 9.5;
- each Extra Package folder shall conform to other provisions as specified in 9.6.

6.3.3.3 Playlists folder

One unique “playlists” folder may be present as a child of the root folder. This folder will be referred to as the Playlists folder in the rest of this document. This folder shall conform to structure provisions specified in 6.7 and other provisions in Clause 12.

6.3.3.4 Root Metadata folder

Exactly one “metadata” folder shall be present as a child of the root folder. This folder will be referred to as the Root Metadata folder in the rest of this document. This folder’s content shall conform to structure provisions specified in 6.5 and other provisions in 11.2.

6.3.3.5 Root Ancillary Data folder

One unique “ancillaryData” folder may be present as a child of the root folder. This folder will be referred to as the Root Ancillary Data folder in the rest of this document. This folder shall conform to structure provisions specified in 6.6 and other provisions in 10.2.

6.3.3.6 Root Checker Reports folder

One unique “checkerReports” folder may be present as a child of the root folder. This folder will be referred to as the Root Checker Reports folder in the rest of this document. This folder shall conform to structure provisions specified in 6.8 and other provisions in Clause 13.

6.4 Subpackage folder structure

6.4.1 Synthetic view (informative)

Subpackage folders contain the following files and subfolders:

- “packingList.xml”;
- (“sound”|“image”|“timedText”|“audiovisual”|“componentized”|“extra”)“Package_”<uuid>;
 - “data”;
 - <subpackage specific filenames>;
 - “metadata”;
 - “techMD_”<uuid>“_0000.xml”;
 - “digiprovMD_”<uuid>“_”<number>“.xml”*;

- ancillaryData"?;
- "ancillaryData_"<uuid>"_<number>."<extension>.

6.4.2 List of existing subpackages

The Image Package, Sound Package, Timed Text Package, Audiovisual Package, Componentized Package, and Extra Package folders share common properties despite different behaviour related to the content described by the prefix. They will be designated as subpackages when referring to common properties.

6.4.3 Subpackage UUID

6.4.3.1 UUID generation

The Unique Universal Identifier shall conform to UUID version 4 according to ISO/IEC 9834-8.

6.4.3.2 UUID presentation

When a UUID appears in a filename, it shall be presented as the plain UUID character sequence, not as an URN.

EXAMPLE "imagePackage_dd779c52-0107-49d4-9a32-6b4ce4cd21f0".

6.4.3.3 UUID consistency

The same UUID shall be used as an abstract subpackage identifier within the Packing List file (see clause 8), and in the subpackage folder name.

6.4.3.4 UUID uniqueness

Considering the content of the asset subfolders – files in the data and ancillaryData folder – in the subpackage folder as a unique set of files, a different set of files shall have a different UUID.

6.4.4 Subpackage folder name

The name of each subpackage folder shall follow the pattern: <prefix>"Package_"<uuid>.

Where:

- <prefix> shall be either "image" for an Image Package, "sound" for a Sound Package, "timedText" for a Timed Text Package, "audiovisual" for an Audiovisual Package, "componentized" for a Componentized Package, or "extra" for an Extra Package;
- <uuid> is the subpackage unique universal identifier of the subpackage as specified in 6.4.3.

6.4.5 Subpackage folder content

6.4.5.1 General

Each subpackage folder shall contain the following file and folders:

- the "packingList.xml" file shall be present and shall conform to provisions specified in 6.4.5.5;
- exactly one "data" folder shall be present. This folder will be referred to as the Subpackage Data folder;
- exactly one "metadata" folder shall be present. This folder will be referred to as the Subpackage Metadata folder;

- one "ancillaryData" folder may be present. This folder will be referred to as the Subpackage Ancillary folder.

6.4.5.2 Subpackage Data folder

The Subpackage Data folder shall contain the data required for each type of subpackage and shall conform to prescriptions listed in Clause 9.

6.4.5.3 Subpackage Metadata folder

6.4.5.3.1 General description

The Subpackage Metadata folder shall contain the metadata required for each type of subpackage and shall conform to prescriptions listed in 11.3.

6.4.5.3.2 Contained files

The Subpackage Metadata folder shall contain zero or one technical metadata file as specified for each kind of subpackage. If present, the name of this file shall be "techMD_"<uuid>"_<technical metadata file number>".xml", where

- <uuid> shall be replaced with the Subpackage UUID described in 6.4.3;
- <technical metadata file number> shall be replaced with "0000".

NOTE The numbering is provisioned for future use, but only one file is present.

The Subpackage Metadata folder may contain one or more provenance metadata files. The name of these files shall be "digiprovMD_"<uuid>"_<provenance metadata file number>".xml", where

- <uuid> shall be replaced with the Subpackage UUID described in 6.4.3;
- <provenance metadata file number> shall be replaced with the next subpackage provenance metadata file number string.

All subpackage provenance metadata filenames shall be consistent, and share the following numbering properties:

- the numbering shall start at 0;
- no distinct subpackage provenance metadata file shall share the same number within the same metadata data folder.
- the decimal number shall be as a string of 4 digits with leading zeros.
- this *provenance metadata file number* shall be incremented for each new subpackage provenance metadata file.

EXAMPLE the first subpackage provenance metadata filename will be "digiprovMD_dd779c52-0107-49d4-9a32-6b4ce4cd21f0_0000.xml", the second one "digiprovMD_dd779c52-0107-49d4-9a32-6b4ce4cd21f0_0001.xml", etc., supposing the Subpackage UUID is "dd779c52-0107-49d4-9a32-6b4ce4cd21f0".

6.4.5.4 Subpackage Ancillary Data folder

6.4.5.4.1 General description

The Ancillary Data folder contained files shall conform to provisions listed in 10.3.

6.4.5.4.2 Contained files

The Ancillary Data folder shall contain one or more ancillary data files.

The name of the files shall be *<ancillary filename>".<extension>*, where:

- *<ancillary filename>* shall be the filename given by the user with alphanumerical ASCII characters, '-' or '_';
- *<extension>* shall be the file extension with alphanumerical ASCII characters.

6.4.5.5 "packingList.xml"

The "packingList.xml" document shall conform to provisions listed in 8.4. The entities references specified in the Packing List shall use the Subpackage UUID as specified in 6.4.3.

6.5 Root Metadata folder structure

6.5.1 General description (informative)

This folder is targeted to convey the root metadata files for the whole package.

6.5.2 Synthetic view (informative)

- "metadata";
 - "descMD_"*<preservation package UUID>"_<root descriptive metadata number>".xml"*;
 - "digiprovMD_"*<preservation package UUID>"_<root provenance metadata number>".xml"*.*.

6.5.3 Root Metadata folder content

The Root Metadata folder shall contain at least one descriptive metadata file, and may contain additional provenance metadata files as specified in 11.2.

6.5.4 Descriptive metadata files

The Root Metadata folder shall contain one or many descriptive metadata files.

The name of these files shall be "descMD_"*<preservation package UUID>"_<root descriptive metadata number>".xml"*, where:

- *<preservation package uuid>* as specified in 6.9;
- *<root descriptive metadata number>* shall be replaced with the next root descriptive metadata number string.

All descriptive metadata filenames shall be consistent, and share the following numbering properties:

- the numbering shall start at 0;
- no distinct descriptive metadata file shall share the same number;
- the decimal number shall be a string of 4 digits with leading zeros;
- the *root descriptive metadata number* shall be incremented for each new descriptive metadata file.

EXAMPLE the first descriptive metadata filename will be “descMD_59c3adeb-e1c3-40a1-8dae-25fef73f15a2_0000.xml”, the second one “descMD_59c3adeb-e1c3-40a1-8dae-25fef73f15a2_0001.xml”, etc., supposing the Preservation Package UUID is “59c3adeb-e1c3-40a1-8dae-25fef73f15a2”.

6.5.5 Provenance metadata files

The Root Metadata folder may contain zero, one or many provenance metadata files.

The name of these files shall be “digiprovMD_”*<preservation package uuid>*“_”*<root provenance metadata number>*“.xml”, where:

- *<preservation package uuid>* is as specified in 6.9;
- *<root provenance metadata number>* shall be replaced with the next root provenance metadata number string.

All provenance metadata filenames shall be consistent, and share the following numbering properties:

- the numbering shall start at 0;
- no distinct provenance metadata file shall share the same number;
- the decimal number shall be a string of 4 digits with leading zeros;
- the *root provenance metadata number* shall be incremented for each new provenance metadata file.

EXAMPLE the first provenance metadata filename will be “digiprovMD_59c3adeb-e1c3-40a1-8dae-25fef73f15a2_0000.xml”, the second one “digiprovMD_59c3adeb-e1c3-40a1-8dae-25fef73f15a2_0001.xml”, etc., supposing the preservation UUID is “59c3adeb-e1c3-40a1-8dae-25fef73f15a2”.

6.6 Root Ancillary Data folder structure

6.6.1 General description (informative)

This folder is targeted to convey documents for human readers related to the whole preservation package.

6.6.2 Synthetic view (informative)

- “ancillaryData”;
- *<filename>*+.

6.6.3 Root Ancillary Data folder content

The Root Ancillary Data folder shall contain at least one root ancillary data file as specified in 10.2.

6.6.4 Root Ancillary Data filename

The name of the files shall be *<root ancillary filename>*“.”*<extension>*, where:

- *<root ancillary filename>* shall be the filename given by the user with alphanumerical ASCII characters, ‘-’ or ‘_’;
- *<extension>* shall be the file extension with alphanumerical ASCII characters.

6.7 Playlists folder structure

6.7.1 General description (informative)

This optional folder is targeted to convey playlist files.

6.7.2 Synthetic view (informative)

- “playlists”;
- “playlist_”<playlist_uuid>“.xml”+.

6.7.3 Playlists folder content

One or more playlist file shall be present in the Playlists folder and shall conform to provisions specified in Clause 12.

6.7.4 Playlist UUID

6.7.4.1 UUID generation

The Unique Universal Identifier shall conform to UUID version 4 according to ISO/IEC 9834-8.

6.7.4.2 UUID presentation

When a UUID appears in a filename, it shall be presented as the plain UUID character sequence, not as an URN.

EXAMPLE “playlist_dd779c52-0107-49d4-9a32-6b4ce4cd21f0”.

6.7.4.3 UUID uniqueness

The UUID shall be the UUID identification element of the playlist within the document, as described in 12.4.

6.7.5 Playlist filename

Each playlist filename shall conform to the pattern “playlist_”<playlist UUID>“.xml”, where:

<playlist UUID> is the playlist UUID specified in 6.7.4.

EXAMPLE a playlist name could be: “playlist_ 9497a702-f3f2-4868-8e82-2535ec85bef8.xml”.

6.8 Root Checker Report folder structure

6.8.1 General description (informative)

This folder contains quality control reports created after the generation of the subpackages.

6.8.2 Synthetic view

- “checkerReports”;
- “checkerReportsList.xml”;
- “reports_”<report process UUID>;
- <report UUID>“-result.”<extension>+.

6.8.3 Root Checker Reports folder content

Exactly one Checker Report List file shall be present. The name of this file shall be “checkerReportsList.xml”. This file shall comply with provisions in 13.2.

One or more Report subfolders shall be present.

The Report subfolder name shall be “reports_”<report process UUID>, where:

- <report process UUID> shall be an UUID representing the checker process.

6.8.4 Content of the Report subfolder

One or more report file shall be present.

6.8.5 Reports filename

The name of the report files shall be <report UUID>“-result.”<extension>, where:

- <report UUID> shall be a new UUID for each new report;
- <extension> shall be the file extension with alphanumerical ASCII characters.

6.9 Preservation Package UUID

6.9.1 UUID generation

The Unique Universal Identifier shall conform to UUID version 4 according to ISO/IEC 9834-8.

6.9.2 UUID presentation

When a UUID appears in a filename, it shall be presented as the plain UUID character sequence, not as an URN.

EXAMPLE “preservationPackage_65346dbc-c39f-4788-b87c-0193d59fff80”.

6.9.3 UUID uniqueness

The UUID shall be the UUID identification element of the Preservation Packing List within the document, as described in 8.3.3.

7 Profiles

7.1 General description (informative)

The Cinema Preservation Package usage is versatile in nature. Some use cases imply the list of acceptable formats stored in subpackages stored to remain open, like content exchange. On the other hand, long term preservation implies a strict selection of well-defined formats. For this reason different profiles are specified.

7.2 Profile attribution

The Cinema Preservation Package shall conform to one of the following profiles:

- the “Interim preservation profile”, to be used when the package content needs to stay flexible;
- the “Long term preservation profile”, to be used when a rigid selection of formats needs to be verified.

The “Interim preservation profile” shall be used as a default. All the recommended formats can be used. Although it is not recommended, other formats may be used as well.

A Cinema Preservation Package instance with the “Long term preservation profile” shall only contain sustainable formats and standards. Clause 9 specifies constraints on subpackages and defines the list of acceptable formats and applicable constraints to comply.

8 Content of Packing List files

8.1 Synthetic view (informative)

The synthetic view of the package is given in Figure 1.

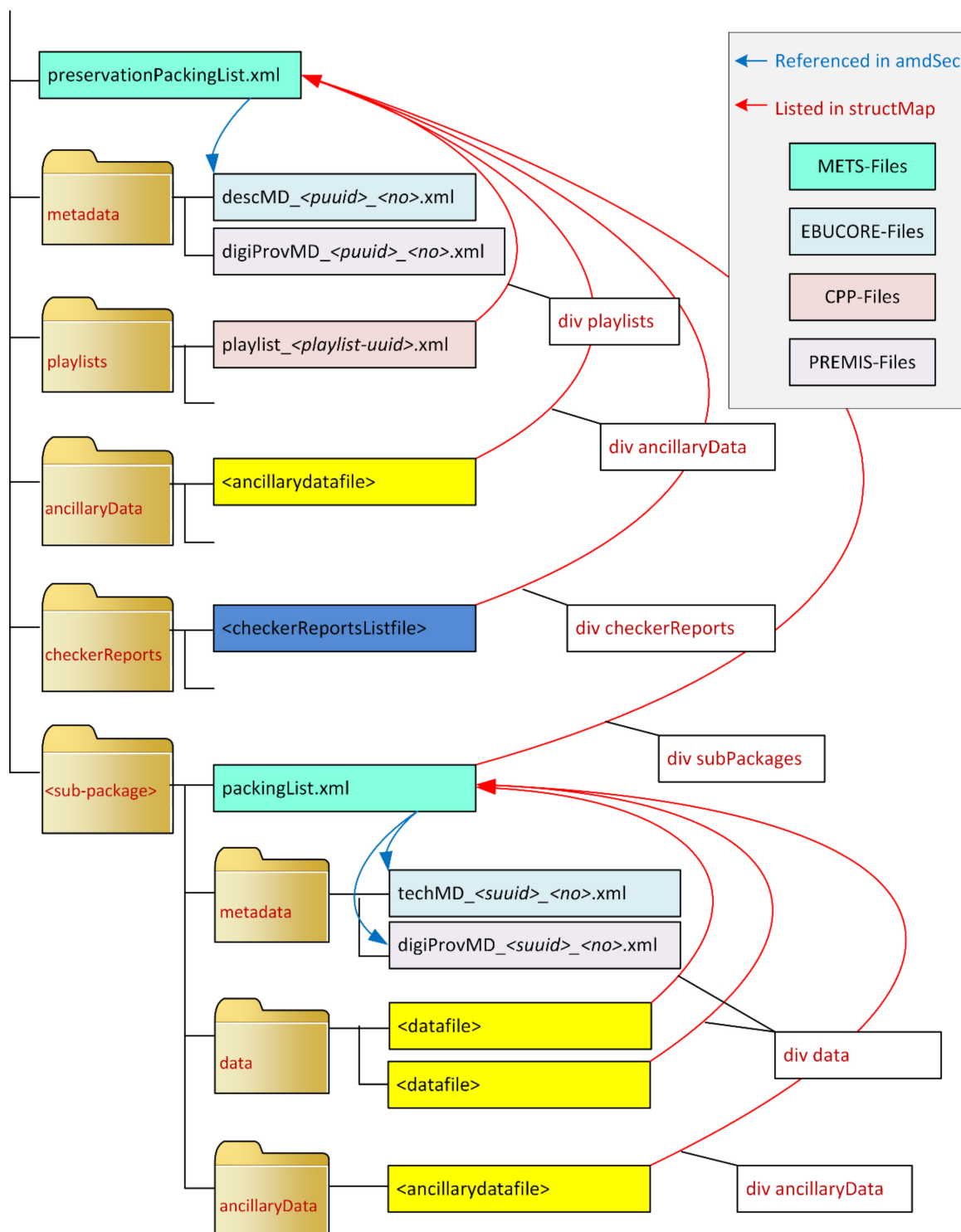


Figure 1 — File references inside Packing List

NOTE “amdSec”, “div” and “structMap” are elements of the METS Schema.

8.2 Packing List

8.2.1 General description (informative)

A Packing List is a METS document containing the following parts, under a root `<mets>` element:

- headers under the < metsHdr > element;
- metadata sections under < dmdSec > and < amdSec > elements;
- list of file pointers with integrity checks attributes under the < fileSec > element;
- definitions of abstract hierarchies under < structMap > elements.

The METS schema allows different elements dedicated to metadata. The metadata provisioned in this document are mapped to the adequate elements.

8.2.2 File conformance

The Packing List file shall conform to the XML standard and follow the METS XML Schema.

8.2.3 Common constraints on Packing List

8.2.3.1 Constraints on < mets > root element

XPath is: “/mets:mets”.

The mets:mets root element shall include the following namespaces:

- “<http://www.loc.gov/METS/>”;
- “<http://www.w3.org/1999/xlink>”;
- “urn:cen.eu:en17650:2022:ns:packingList” as specified in Annex A.

The following prose in this section assume that the used prefix for these namespaces are respectively “mets”, “xlink” and “cpp”, elements without namespaces belongs to the Mets schema.

Under the < mets:mets > root element, the elements in Table 1 shall be present.

Table 1 — Mandatory elements on the < mets > root element

Category	Topic	METS element	Element Arity	Attribute or subelement	Sub-element mandatory	value
Headers	Header data	metsHdr	1	@CREATEDATE	x	<i>creation date in ISO 8601-1 format</i>
				@cpp:packingList	x	<i>packingListKind</i>
				@cpp:packageProfile	o	<i>packageProfile</i>
				@cpp:packageFormatVersion	o	<i>packageFormatVersion</i>
				#children	x	See 8.2.3.2
Metadata	Descriptive metadata	dmdSec	0..n	@ID	x	<i>descriptiveMetadataId</i>
				#children	x	For Preservation Packing List see 8.3.4.2
	Provenance metadata	amdSec	1	@ID	x	<i>administrativeMetadataId</i>
				#children	x	For Preservation Packing List see 8.3.4.3, for Subpackage Packing List see 8.4.4.2

Category	Topic	METS element	Element Arity	Attribute or subelement	Sub-element mandatory	value
Files management	File section	fileSec	1	#children	x	For Preservation Packing List see 8.3.4.4, for Subpackage Packing List see 0
Abstract trees	Package physical description	structMap	1	@TYPE	x	"physical"
				#children	x	For Preservation Packing List see 8.3.4.9, for Subpackage Packing List see 8.4.4.6
	Package logical description	structMap	0..n	@TYPE	x	"logical"
				#children	x	For Preservation Packing List see 8.3.4.15.

See specific constraints for Preservation Packing List in 8.3.4.1 and for Subpackage Packing List in 8.4.4.1.

8.2.3.2 Constraints on the <metsHdr> element

XPath is: "/mets:mets/mets:metsHdr/".

The elements in Table 2 shall be present.

Table 2— Mandatory elements on the <metsHdr> element

Topic	METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
Agent	agent	1	@ROLE	x	"CREATOR"
			@TYPE	x	"OTHER"
			#children	x	See Table 3

The elements in Table 3 shall be present.

Table 3 — Mandatory elements on the <agent> element

Topic	METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
Name	name	1	#text	x	<i>packager agent name</i>

packager agent name shall be the name of the person or institution who produced the CPP package.

8.2.3.3 <fileGrp> elements common constraints

XPath is: "/mets:mets/mets:fileSec/mets:fileGrp/".

The element listed in Table 4 shall be repeated for each file belonging to the <fileGrp>.

Table 4 — mandatory element for all <fileGrp> elements

METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
file	1..n	@ID	x	<i>fileIdentifier</i>
		@MIMETYPE	x	<i>mime type</i>
		@SIZE	x	<i>file size</i>
		@CHECKSUM	x	<i>file checksum</i>
		@CHECKSUMTYPE	x	<i>checksum algorithm</i>
		#children	x	See Table 5

fileIdentifier shall be replaced as specified in the next sections.

mime type shall be replaced as specified in the next sections.

file size shall be an integer value representing the number of bytes of the file.

checksum algorithm shall be one of the values enumerated by METS.

file checksum shall be the hash value of the file computed with the *checksum algorithm*.

The element in Table 5 shall be present.

Table 5 — Mandatory element of the <file> element

METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
FLocat	1	@LOCTYPE	x	"URL"
		@xlink:type	x	"simple"
		@xlink:href	x	<i>relative path to file</i>

relative path to file shall be replaced as specified in the next sections.

8.3 Preservation Packing List

8.3.1 General description (informative)

The Playlist files and Root Ancillary Data files are listed in <fileSec> element and referenced in <structMap> elements with the identifiers specified in the <fileSec> element.

The Subpackage Packing List files are listed as files in the <fileSec> element and appear in the <structMap> element as extension of the "physical" abstract tree.

8.3.2 Files to be referenced

The Preservation Packing List shall list all the following files:

- all files in the Playlists folder;
- all files in the Root Ancillary Data folder;
- all files in the Root Metadata folder;

- all Subpackage Packing List files from the Subpackage folders;
- the Checker Report List file from the Checker Reports folder.

The Preservation Packing List content shall always reflect the name, path, size and content hashes of the actual files.

8.3.3 Preservation Package Unique Identifier uniqueness

Each Preservation Packing List unique content shall have a distinct package identifier (UUID).

NOTE 1 The names of the Root Metadata files will change accordingly.

NOTE 2 Because this preservation package identifier identifies the whole mastered package, and not only the Preservation Packing List, the filenames of the associated root metadata use this identifier.

8.3.4 Specific constraints on the Preservation Packing List

8.3.4.1 Constraints on the <mets> root element

packingListKind shall be "preservationPackingList".

packageProfile shall be present and shall be "longTermPreservationProfile" if the subpackages are all compliant with the Long term preservation profile, or "interimPreservationProfile" if not.

packageFormatVersion shall be present, and shall be "urn:cen.eu:en17650:2022:ns:cpp#release".

NOTE This value is meant to change with each revision of this standard.

The *dmdSec* element shall be repeated once for each Descriptive Metadata file in the Root Metadata folder.

The *descriptiveMetadataId* shall be: "descMD_"<uuid>"_<number>", where:

- <uuid> shall be replaced with the Preservation Package UUID as specified in 6.9;
- <number> shall be the number of the file.

The <uuid> and <number> values shall be the UUID and number in the name of the file.

EXAMPLE A possible value is "descMD_69b0eccf-373c-454a-8b42-d53701af8898_0000".

NOTE The *descriptiveMetadataId* is similar to the targeted filename without extension.

The *administrativeMetadataId* shall be "amdMD_"<uuid>, where:

- <uuid> shall be replaced with the Preservation Package UUID as specified in 6.9.

8.3.4.2 Constraints on the <dmdSec> element

XPath is: "/mets:mets/mets:dmdSec/".

The element listed in Table 6 shall be present.

Table 6 — Mandatory element of the <dmdSec> element

Topic	METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
Descriptive metadata	mdRef	1	@LOCTYPE	x	"URL"
			@xlink:href	x	<i>relative path to file</i>
			@MDTYPE	x	"OTHER"
			@OTHERMDTYPE	x	"EBUCORE"
			@MIMETYPE	x	"text/xml"
			@SIZE	x	<i>file size</i>
			@CHECKSUM	x	<i>file checksum</i>
			@CHECKSUMTYPE	x	<i>checksum algorithm</i>

relative path to file shall be the path from the root folder to the Descriptive Metadata file including the filename.

file size shall be an integer value of the number of bytes of the file.

checksum algorithm shall be one of the values enumerated by METS.

file checksum shall be the hash value of the file computed with the checksum algorithm.

8.3.4.3 Constraints on the <amdSec> element

XPath is: "/mets:mets/mets:amdSec".

The element listed in Table 7 shall be present.

Table 7 — Mandatory elements of the <amdSec> element

Topic	METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
Provenance metadata	digiprovMD	0..n	@ID	x	<i>digiprovMDId</i>
			#children	x	See Table 8

The <digiprovMD> element shall be repeated once for each Root Provenance file in the Root Metadata folder.

The *digiprovMDId* shall be "digiprovMD_"<uuid>"_ "<number>, where:

- <uuid> shall be replaced with the Preservation Package UUID as specified in 6.9.
- <number> shall be the number of the file.

The <uuid> and <number> sections shall be the UUID and number of the digital provenance filename.

The element listed in Table 8 shall be present in each occurrence of the <digiprovMD> element.

Table 8 — Mandatory element of the <digiprovMD> element

Topic	METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
digiprovMD	mdRef	1	@LOCTYPE	x	"URL"
			@xlink:href	x	<i>relative path to file</i>
			@MDTYPE	x	"PREMIS"
			@MIMETYPE	x	"text/xml"
			@SIZE	x	<i>file size</i>
			@CHECKSUM	x	<i>file checksum</i>
			@CHECKSUMTYPE	x	<i>checksum algorithm</i>

relative path to file shall be the path from the root folder to the Digital Provenance Metadata file including the filename.

file size shall be an integer value of the number of bytes of the file.

checksum algorithm shall be one of the values enumerated by METS.

file checksum shall be the hash value of the file computed with the checksum algorithm.

8.3.4.4 Constraints on the <fileSec> element

XPath is: "/mets:mets/mets:fileSec".

The elements listed in Table 9 shall be present.

Table 9 — Mandatory elements of the <fileSec> element

Category	Topic	METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
File management	Ancillary files	fileGrp	1	@ID	x	<i>ancillaryFileGrpId</i>
				#children	x	See 8.3.4.5
	Playlist files	fileGrp	1	@ID	x	<i>playlistFileGrpId</i>
				#children	x	See 8.3.4.6
	CheckerReports files	fileGrp	0..1	@ID	x	<i>checkerReportsFileGrpId</i>
				#children	x	See 8.3.4.7
	Subpackages	fileGrp	1..n	@ID	x	<i>subpackageFileGrpId</i>
				#children	x	See 8.3.4.8

ancillaryFileGrpId shall be "ancillaryData_"<uuid>"_fileGrp", where:

— <uuid> shall be the Preservation Package UUID as specified in 6.9.

playlistFileGrpId shall be "playlists_"<uuid>"_fileGrp", where:

— <uuid> shall be the Preservation Package UUID as specified in 6.9.

checkerReportsFileGrpId shall be "checkerReports_"<uuid>"_fileGrp", where:

- *<uuid>* shall be the Preservation Package UUID as specified in 6.9.

In addition to the Ancillary files and Playlist files *fileGrp* elements, the *<fileGrp>* element shall be repeated once for each Packing List file present in subpackage folders.

For the subpackages *<fileGrp>* elements the value of the subpackageFileGrpId identifier shall be: *<prefix>“_”<uuid>“_fileGrp”*, where:

- *<prefix>* is the subpackage prefix as specified in 6.4.4;
- *<uuid>* is the subpackage unique identifier as specified in 6.4.3.

8.3.4.5 Ancillary files *<fileGrp>* element constraints

The *<fileGrp>* element shall conform to 8.2.3.3.

fileIdentifier shall be “ancillaryData_”*<uuid>*“_”*<number>*, where:

- *<uuid>* shall be the Preservation Package UUID as specified in 6.9.
- *<number>* shall be the order of the file in the Root Ancillary Data folder starting at 0 as a string of 4 digits with leading zeros, using lexicographic order.

EXAMPLE “ancillaryData_69b0eccf-373c-454a-8b42-d53701af8898_0000”.

mime type corresponds to the registered mime type of the file at IANA Media Types, e.g. “application/pdf” or “image/tiff”.

relative path to file shall be the path to the ancillary file including the filename.

8.3.4.6 Playlist files *<fileGrp>* element constraints

The *<fileGrp>* element shall conform to 8.2.3.3.

fileIdentifier shall be “playlist_”*<uuid>*“_”*<number>*, where:

- *<uuid>* shall be the Preservation Package UUID as specified in 6.9;
- *<number>* shall be the order of the file in the Playlists folder starting at 0 as a string of 4 digits with leading zeros, using lexicographic order.

EXAMPLE “playlist_69b0eccf-373c-454a-8b42-d53701af8898_0000”.

mimetype shall be “text/xml”

relative path to file shall be the path to the Playlist file including the filename.

8.3.4.7 CheckerReportsList file *<fileGrp>* element constraints

The *<fileGrp>* element shall conform to 8.2.3.3.

fileIdentifier shall be “checkerReports_”*<uuid>*“_0000”, where:

- *<uuid>* shall be the Preservation Package UUID as specified in 6.9.

EXAMPLE “checkerReports_ bfe35792-4de2-49bd-9906-52637cd46ed0_0000”

mimetype shall be “text/xml”

relative path to file shall be the path to the CheckerReportList file, in the Root Checker Reports folder, including the filename.

8.3.4.8 Subpackages <fileGrp> element constraints

The <fileGrp> element shall conform to 8.2.3.3.

fileIdentifier shall be <prefix>“_”<uuid>“_packingList”, where:

- <prefix> shall reflect the type of package prefix as specified in 6.4.4;
- <uuid> shall reflect the UUID of the subpackage as specified in 6.4.3.

mime type shall be “text/xml”

relative path to file shall be the path to the Packing List file including the filename.

8.3.4.9 Constraints on the “physical” <structMap> element

XPath is: “/mets:mets/mets:structMap[@TYPE=“physical”]”.

The element listed in Table 10 shall be present.

Table 10 — Mandatory elements in the <structMap> element

Topic	METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
Abstract physical tree root	div	1	@ID	x	<i>preservationPackageId</i>
			@TYPE	x	“CPP-Package”
			@DMDID	x	<i>descriptiveMetadataId</i>
			@LABEL	x	<i>label</i>
			@ADMID	o	<i>administrativeMetadataId</i>
			#children	x	See 8.3.4.10

preservationPackageId shall be “preservationPackage_”<uuid>, where:

- <uuid> shall be the Preservation Package UUID as specified in 6.9.

EXAMPLE “preservationPackage_65346dbc-c39f-4788-b87c-0193d59fff80”.

descriptiveMetadataId and *administrativeMetadataId* are specified in 8.3.4.1

label shall be the name of the root folder name as specified in 6.3.2.

8.3.4.10 Constraints to the “physical” <structMap> sub <div>

XPath is: “/mets:mets/mets:structMap[@TYPE=“physical”]/mets:div”.

The element listed in Table 11 shall be present.

Table 11 — Mandatory elements in the “physical” <div> element

Topic	METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
Ancillary data	div	1	@TYPE	x	“ancillaryData”
			#children	x	See 0
Playlist	div	1	@TYPE	x	“playlists”
			#children	x	See 8.3.4.12
Subpackages	div	1	@TYPE	x	“subPackages”
			#children	x	See 8.3.4.13
Checker reports	div	1	@TYPE	x	“checkerReports”
			#children	x	See 8.3.4.14

8.3.4.11 “ancillaryData” <div>

XPath is:

“/mets:mets/mets:structMap[@TYPE="physical"]/mets:div /mets:div[@TYPE="ancillaryData"]”.

The element listed in Table 12 shall be present as many times as the number of items present in the Ancillary Data fileGrp.

Table 12 — Mandatory in the ancillary data <div> element

METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
fptr	1	@FILEID	x	<i>ancillaryFileId</i>

ancillaryFileId shall be the fileIdentifier as specified in 8.3.4.5.

8.3.4.12 “playlists” <div>

XPath is:

“/mets:mets/mets:structMap[@TYPE="physical"]/mets:div /mets:div[@TYPE="playlists"]”.

The element listed in Table 13 shall be present as many times as the number of items present in the Playlist fileGrp.

Table 13 — Mandatory element in the “playlist” <div> element

METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
fptr	1..n	@FILEID	x	<i>playlistFileId</i>

For each playlist file *playlistFileId* shall be the *fileIdentifier* as specified in 8.3.4.6.

8.3.4.13 “subPackages” <div>

XPath is:

“/mets:mets/mets:structMap[@TYPE="physical"]/mets:div/mets:div[@TYPE="subPackages"]”.

The element listed in Table 14 shall be present as many times as subpackage folders with their relative Packing List.

Table 14 — Mandatory element in the subpackage <div> element

METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
mptr	1..n	@LOCTYPE	x	"URL"
		@xlink:type	x	"simple"
		@xlink:href	x	<i>relative path to file</i>

relative path to file shall be the path to the Subpackage Packing List file including the filename.

8.3.4.14 "checkerReports" <div>

XPath is:

`"/mets:mets/mets:structMap[@TYPE="physical"]/mets:div/mets:div[@TYPE="checkerReports"]`

The element listed in Table 15 shall be present.

Table 15 — Mandatory element in the subpackage <div> element

METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
fptr	1	@FILEID	x	<i>checkerReportsFileId</i>

The *checkerReportsFileId* shall be the *fileIdentifier* as specified in 8.3.4.7.

8.3.4.15 Constraints to the "logical" <structMap>

XPath is: `"/mets:mets/mets:structMap[@TYPE="logical"]`.

The element listed in Table 16 shall be present.

Table 16 — Mandatory element in the "logical" <structMap> element

METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
div	1	@DMDID	x	<i>descriptiveMetadataId</i>
		@LABEL	x	<i>Version name</i>
		#children	x	See 8.3.4.16

descriptiveMetadataId shall be: "descMD_"<uuid>"_ "<number>, where:

- <uuid> shall be replaced with the Preservation Package UUID as specified in 6.9;
 - <number> shall be the number of the descriptive metadata file associated with the current version.
- version name* shall be the name of the version.

8.3.4.16 Constraints to the “logical” <structMap> sub <div>

XPath is: “/mets:mets/mets:structMap[@TYPE="logical"]/div”.

The element listed in Table 17 shall be present as many times as subpackages related to the current version.

Table 17 — Mandatory element in the subpackage <div> element

METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
mptr	1..n	@LOCTYPE	x	“URL”
		@xlink:type	x	“simple”
		@xlink:href	x	<i>relative path to file</i>

relative path to file shall be the path to the Subpackage Packing List file including the filename.

8.4 Subpackage Packing List

8.4.1 General description (informative)

The Data files and Root Ancillary Data files are listed in <fileSec> and reflected in the abstract structure under <structMap> with the identifiers specified in the <fileSec>.

8.4.2 Files to be referenced

The Subpackage Packing List shall list all the following files:

- all files in the Subpackage Data folder;
- all files in the Subpackage Ancillary Data folder;
- all files in the Subpackage Metadata folder.

8.4.3 Subpackage Unique Identifier uniqueness

The Subpackage Packing List content shall always reflect the name, path, size and content hashes of the actual files.

Each Subpackage Packing List unique content shall have a distinct subpackage package identifier (UUID).

8.4.4 Constraints on the Subpackage Packing List

8.4.4.1 Constraints on the <mets> root element

packingListKind shall be “subpackagePackingList”

packageProfile shall not be present.

packageFormatVersion shall not be present.

The *dmdSec* element shall not be present.

The *administrativeMetadataId* shall be “amdMD_”<uuid>, where:

- <uuid> shall be replaced with the Subpackage UUID as specified in 6.4.3.

8.4.4.2 Constraints on the <amdSec> element

XPath is: “/mets:mets/mets:amdSec”.

The elements listed in Table 18 shall be present.

Table 18 — Mandatory elements of the <amdSec> element

Topic	METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
Technical metadata	techMD	1..n	@ID	x	<i>techMDId</i>
			#children	x	See Table 19
Provenance metadata	digiprovMD	0..n	@ID	x	<i>digiprovMDId</i>
			#children	x	See Table 20

The <techMD> element shall be repeated once for each subpackage technical metadata files present in the “metadata” folder.

techMDId shall be “techMD_”<uuid>“_”<number>, where:

- <uuid> shall be replaced with the Subpackage UUID as specified in 6.4.3;
- <number> shall be the number of the file.

The <uuid> and <number> section shall reflect the UUID and number of the technical metadata filename.

The element listed in Table 19 shall be present in each occurrence of the <techMD> element.

Table 19 — Mandatory element of the <techMD> element

Topic	METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
techMD	mdRef	1	@LOCTYPE	x	“URL”
			@xlink:href	x	<i>relative path to file</i>
			@MDTYPE	x	“OTHER”
			@OTHERMDTYPE	x	“EBUCORE”
			@MIMETYPE	x	“text/xml”
			@SIZE	x	<i>file size</i>
			@CHECKSUM	x	<i>file checksum</i>
			@CHECKSUMTYPE	x	<i>checksum algorithm</i>

The <digiprovMD> element shall be repeated once for each subpackage digital provenance files in the “metadata” folder.

digiprovMDId shall be “digiprovMD_”<uuid>“_”<number>, where:

- <uuid> shall be the subpackage UUID as specified in 6.4.3;
- <number> shall be the number of the file.

The <uuid> and <number> section shall reflect the UUID and number of the digital provenance filename.

The element listed in Table 20 shall be present in each occurrence of the <digiprovMD> element.

Table 20 — Mandatory element of the <digiprovMD> element

Topic	METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
digiprovMD	mdRef	1	@LOCTYPE	x	"URL"
			@xlink:href	x	<i>relative path to file</i>
			@MDTYPE	x	"PREMIS"
			@MIMETYPE	x	"text/xml"
			@SIZE	x	<i>file size</i>
			@CHECKSUM	x	<i>file checksum</i>
			@CHECKSUMTYPE	x	<i>checksum algorithm</i>

8.4.4.3 Constraints on the <fileSec> element

XPath is: "/mets:mets/mets:fileSec".

The elements listed in Table 21 shall be present.

Table 21 — mandatory elements of the fileSec element

Topic	METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
Ancillary files	fileGrp	0..1	@ID	x	<i>ancillaryFileGrpId</i>
			#children	x	See 8.4.4.4
Data	fileGrp	1..n	@ID	x	<i>dataFileGrpId</i>
			#children	x	See 8.4.4.5

ancillaryFileGrpId shall be "ancillaryData_"<uuid>"_fileGrp", where:

- <uuid> shall be the subpackage UUID as specified in 6.4.3.

dataFileGrpId shall be "data_"<uuid>"_fileGrp", where:

- <uuid> shall be the subpackage UUID as specified in 6.4.3.

8.4.4.4 Ancillary files <fileGrp> element constraints

The <fileGrp> element shall conform to 8.2.3.3.

fileIdentifier shall be "ancillaryData_"<uuid>"_"<number>, where:

- <uuid> shall be the Subpackage UUID as specified in 6.9;
- <number> shall be the order of the file in the Root Ancillary folder starting at 0 as a string of 4 digits with leading zeros, using lexicographic order.

EXAMPLE "ancillaryData_69b0eccf-373c-454a-8b42-d53701af8898_0000"

mime type shall be “application/pdf” or “image/tiff” according to the file type.

relative path to file shall be the path to the ancillary file including the filename.

8.4.4.5 Data <fileGrp> element constraints

The <fileGrp> element shall conform to 8.2.3.3.

fileIdentifier shall be “data_”<uuid>“_”<number>, where:

- <uuid> shall be the Subpackage UUID as specified in 6.9;
- <number> shall reflect the number of the file.

mime type shall be the MIME type

relative path to file shall be the path to the file including the filename.

8.4.4.6 Constraints on the “physical” <structMap> element

XPath is: “/mets:mets/mets:structMap[@TYPE=“physical”]”.

The element listed in Table 22 shall be present.

Table 22 — mandatory elements in the <structMap> element

Topic	METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
Abstract physical tree root	div	1	@ID	x	<i>subpackageId</i>
			@LABEL	x	<i>subpackageLabel</i>
			@TYPE	x	<i>subpackageType</i>
			#children	x	See 8.4.4.7

subpackageId shall be “subPackage_”<uuid>, where:

- <uuid> shall be the Subpackage UUID as specified in 6.4.3.

subpackageLabel shall be <typePrefix>“_”<uuid>, where:

- <typePrefix> shall be the prefix denoting the type of package as specified in 6.4.4;
- <uuid> shall be the Subpackage UUID as specified in 6.4.3.

EXAMPLE “imagePackage_6aac082c-008f-4827-becb-a4697d2383a7”.

subpackageType shall be <typePrefix>, where:

- <typePrefix> shall be the prefix denoting the type of package as specified in 6.4.4.

8.4.4.7 Constraints to the “physical” <structMap> sub <div>

XPath is: “/mets:mets/mets:structMap[@TYPE=“physical”]/mets:div”.

The elements listed in Table 23 shall be present.

Table 23 — Mandatory elements in the “physical” <div> element

Topic	METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
Ancillary data	div	0..1	@TYPE	x	“ancillaryData”
			@ID	x	<i>ancillaryDataDivId</i>
			#children	x	See 8.4.4.8
Data	div	1	@TYPE	x	“data”
			@ID	x	<i>dataDivId</i>
			@ADMID	x	<i>administrativeMetadataId</i>
			#children	x	See 8.4.4.9

ancillaryDataDivId shall be “ancillaryData_”<uuid>, where:

- <uuid> shall be the Subpackage UUID as specified in 6.4.3.

dataDivId shall be “data_”<uuid>, where:

- <uuid> shall be the Subpackage UUID as specified in 6.4.3.

administrativeMetadataId is specified in 8.3.4.1.

8.4.4.8 Ancillary data <div>

XPath is: “/mets:mets/mets:structMap/mets:div[@TYPE=“ancillaryData”]”.

The element listed in Table 24 shall be present as many times as the number of items present in the Ancillary Data <fileGrp>.

Table 24 — Mandatory element in the ancillary data div element

METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
fptr	1..n	@FILEID	x	<i>ancillaryFileId</i>

ancillaryFileId shall be the fileIdentifier as specified in 8.4.4.4.

8.4.4.9 Data <div>

XPath is: “/mets:mets/mets:structMap/mets:div[@TYPE=“data”]”.

The element listed in Table 25 shall be present as many times as the number of items present in the Data <fileGrp>.

Table 25 — Mandatory element in the data div element

METS element	Element Arity	Attribute or subelement	Subelement mandatory	value
fptr	1..n	@FILEID	x	<i>dataFileId</i>

dataFileId shall be the fileIdentifier as specified in 8.4.4.5.

9 Specification of the subpackages

9.1 Image Package

9.1.1 General description

The Image Package is intended to convey image sequences.

9.1.2 General provisions

The Image Package folder shall follow the provisions listed in 6.4.

9.1.3 Data folder content

The Data folder shall contain at least one image file.

9.1.4 Sequence homogeneity

Within the Data folder, all the images shall be specified with the same standard or specification, with the same image dimensions, the same bitdepth, the same colourspace, the same number of channels.

9.1.5 Sequence continuity

The Data folder shall contain only continuous images sequences.

In the case of missing parts in a reel, the images of the original work item shall be split in as much image subpackages as continuous parts.

9.1.6 Image filenames

The name of the images files shall follow the pattern: *<image prefix>“_”<imageNumber>“.”<extension>*

Where:

- *<image prefix>* is an image sequence identifier chosen by the preservation facility providing the sequence. Only alphanumerical ASCII characters and ‘-’ shall be used. The same image prefix shall be used for all images of the same sequence;
- *<imageNumber>* is the image number and should be a string of 8 digits with leading zeros;
- *<extension>* is the file extension of the given image standard.

NOTE 1 DPX files usually have a “dpx” extension, TIFF files a “tif” or “tiff” extension, OpenEXR files an “exr” extension.

NOTE 2 If not compliant, original filenames are not retained, but it is possible to store them in provenance metadata using PREMIS syntax.

9.1.7 Image numbering

Image File sequences may start at any non-negative number and the numbers shall be contiguous.

9.1.8 Image Package standard conformance

9.1.8.1 List of standards

If the subpackage belongs to a Cinema Preservation Package with the “Long term preservation profile”, then all images contained in an Image Package Data folder, shall belong to one of the standards marked as allowed in the corresponding column in Table 26.

Else, for the “Interim preservation profile”, all images contained in an Image Package Data folder, should belong to one of the standards marked as allowed in the corresponding column in Table 26.

Table 26 — Image Package referenced standards

Standards	Short designation		Interim preservation profile	Long term preservation profile	Additional constraints section
	LoC	CPP			
SMPTE ST 268	“DPX_2”		allowed	allowed	
ISO 12234-2	“TIFF_6”		allowed	allowed	
OpenEXR		“OPENEXR”	allowed	not allowed	
SMPTE ST 2065-4		“ACES”	allowed	allowed	
ISO/IEC 10918-*	“JPEG”		allowed	allowed	
ISO/IEC 15948	“PNG”		allowed	allowed	
ISO/IEC 15444-1 (JPEG 2000)	“JP2_FF”		allowed	allowed	9.1.8.2

9.1.8.2 Constraints on JPEG 2000

Only the cinema profiles, archive profiles, broadcast profiles or IMF profiles specified by the Rsiz parameter in Table 27 and as defined in ISO/IEC 15444-1 shall be used.

Table 27 — JPEG 2000 profiles

Profile Type	Rsiz Value (bits)		Profile Name
	MSB	LSB	
Cinema	0000 0000 0000 0011		2k Digital Cinema Profile
Cinema	0000 0000 0000 0100		4k Digital Cinema Profile
Archive	0000 0000 0000 0111		Long-term storage Profile
Broadcast	0000 0001 0000 xxxx		Broadcast Contribution Single Tile Profile
Broadcast	0000 0010 0000 xxxx		Broadcast Contribution Multi-tile Profile
Broadcast	0000 0011 0000 0110		Broadcast Contribution Multi-tile Reversible Profile
Broadcast	0000 0011 0000 0111		Broadcast Contribution Multi-tile Reversible Profile
IMF	0000 0100 yyyy xxxx		2k IMF Single Tile Lossy Profile
IMF	0000 0101 yyyy xxxx		4k IMF Single Tile Lossy Profile
IMF	0000 0110 yyyy xxxx		8k IMF Single Tile Lossy Profile
IMF	0000 0111 yyyy xxxx		2k IMF Single/Multi Tile Reversible Profile
IMF	0000 1000 yyyy xxxx		4k IMF Single/Multi Tile Reversible Profile
IMF	0000 1001 yyyy xxxx		8k IMF Single/Multi Tile Reversible Profile

NOTE The bits identified as xxxx in Table 27 describe Mainlevel. The bits identified as yyyy in Table 27 describe Sublevel.

9.1.9 Technical Metadata

Exactly one technical metadata file shall be present.

The technical details of the data files shall be documented in the metadata file as specified in 11.3.

9.2 Sound Package

9.2.1 General description

The Sound Package is intended to convey audio tracks.

9.2.2 General provisions

The Sound Package folder shall follow the provisions listed in 6.4.

9.2.3 Data folder content

The Data folder shall contain at least one sound file.

9.2.4 Soundfield consistency

All the sound files in the Data folder should belong to the same soundfield and to the same sound sequence.

In case of a multichannel sound field, as many files as channels shall be present in the folder.

Each sound file in the Data folder shall contain exactly one audio channel.

Different soundfields shall be stored in different separate sound packages. For instance, audio description files may not be stored in the same package as main audio files. Relations between elements can be documented in the playlist.

9.2.5 Homogeneity

All sound files within the Data folder shall follow the same standards, have the same sampling rate, bit depth, and duration, and shall only contain one track per file.

9.2.6 Time representation consistency

The tracks into the different files shall be mutually synchronised. A simultaneous playback of the files shall lead to a time consistent soundfield.

9.2.7 Sound file names

The name of the sound files shall follow the pattern: *<sound file prefix>["_ "<channel>"]".<extension>*.

Where:

- *<sound file prefix>* shall be a sound sequence identifier chosen by the preservation facility providing the sequence. Only alphanumerical ASCII characters and '-' shall be used. The same sound filename shall be used for all sound files of the same sequence;
- *["_ "<channel>]* part may be present and if present, *<channel>* shall be a channel number as a string of two digits with a leading zero;
- *<extension>* is the file extension of the sound container standard used.

NOTE If not compliant, original filenames are not retained, but it is possible to store them in provenance metadata using PREMIS syntax.

9.2.8 Sound Package Standard conformance

If the subpackage belongs to a Cinema Preservation Package with the “Long term preservation profile”, then all sound files contained in a Sound Package Data folder, shall belong to one of the standards marked as allowed in the corresponding column in Table 28.

Else, for the “Interim preservation profile”, all sound files contained in a Sound Package Data folder, should belong to one of the standards marked as allowed in the corresponding column in Table 28.

Table 28 — Sound Packages referenced standards

Standards	Short designation		Interim preservation profile	Long term preservation profile	Additional constraints section
	LoC	CPP			
RIFF Wave	“WAVE”		allowed	allowed	
ITU-R BS.1352-3	“WAVE_BWF_1”		allowed	allowed	
ITU-R BS.2088	“WAVE_BWF_2”		allowed	allowed	
Flac		“FLAC”	allowed	not allowed	
ISO/IEC 14496-3	“AAC_MP4”		allowed	allowed	
ETSI TS 102 366	“AC-3_A”		allowed	allowed	
ISO/IEC 13818-3	“MP3_FF”		allowed	allowed	

9.2.9 Technical Metadata

Exactly one technical metadata file shall be present.

The technical specifics of the data files shall be documented in the metadata file as specified in 11.3.

9.3 Timed Text Package

9.3.1 General description

The Timed Text Package is intended to convey timed text files such as subtitles, captions, etc.

9.3.2 General provisions

The Timed Text Package folder shall follow the provisions listed in 6.4.

9.3.3 Data folder content

The Data folder shall contain at least one timed text file.

The timed text file may contain references to font files or subpicture images files. In this case, the referenced files shall be included in a subfolder structure according to the timed text specification.

9.3.4 Timed text filename

The name of the timed text files shall follow the pattern: *<timed text filename>“.”<extension>*.

Where:

- *<timed text filename>* shall be an identifier chosen by the preservation facility providing the subpackage. Only alphanumerical ASCII characters and ‘-’ shall be used. The same timed text filename shall be used for all subpicture images of the subpackage, if any.

— *<extension>* is the file extension of the given Timed Text standard.

NOTE If not compliant, original filenames are not retained, but it is possible to store them in provenance metadata using PREMIS syntax.

9.3.5 Technical Metadata

Exactly one technical metadata file shall be present.

The technical specifics of the data files shall be documented in the metadata file as specified in 11.3.

9.3.6 Constraints

9.3.6.1 Character encoding

All text files, including xml files shall be encoded in UTF-8.

9.3.7 Timed Text Package Standard conformance

If the subpackage belongs to a Cinema Preservation Package with the “Long term preservation profile”, then all the timed text files contained in a Timed Text Package Data folder, shall belong to one of the standards marked as allowed in the corresponding column in Table 29.

Else, for the “Interim preservation profile”, all timed text files contained in a Timed Text Package Data folder, should belong to one of the standards marked as allowed in the corresponding column in Table 29.

Table 29 —timed text subpackages referenced standards

Standard	Short designation		Interim preservation profile	Long term preservation profile	Additional constraints section
	LoC	CPP			
W3C ttml-ims1.1		“IMSC1.1”	allowed	allowed	
SMPTE ST 428-7		“DCDM_Subtitle”	allowed	allowed	
ISO/IEC 15948	“PNG”		allowed	allowed	
EBU TECH 3264-E		“STL”	allowed	allowed	
EBU TECH 3350		“EBU-TT”	allowed	allowed	

9.4 Audiovisual Package

9.4.1 General description

The Audiovisual Package is intended to convey files embedding image tracks, sound tracks or subtitles into the same one file.

9.4.2 General provisions

The Audiovisual Package folder shall follow the provisions listed in 6.4.

9.4.3 Data folder content

The Data folder shall contain only one audio-visual file.

NOTE The file contents can be image, audio, or timed text only, or any combination thereof.

9.4.4 Audiovisual filename

The name of the audiovisual files shall follow the pattern: *<audiovisual filename>".<extension>.*

Where:

- *<audiovisual filename>* shall be an identifier chosen by the preservation facility providing the subpackage. Only alphanumerical ASCII characters and '-' shall be used;
- *<extension>* is the file extension of the given audiovisual standard, see Table 30.

NOTE If not compliant, original filenames are not retained, but it is possible to store them in provenance metadata using PREMIS syntax.

9.4.5 Audiovisual Packages standard conformance

9.4.5.1 Containers

If the subpackage belongs to a Cinema Preservation Package with the "Long term preservation profile", then all audiovisual containers files contained in an Audiovisual Package Data folder, shall belong to one of the standards marked as allowed in the corresponding column in Table 30.

Else, for the "Interim preservation profile", all audiovisual containers files contained in an Audiovisual Package Data folder, should belong to one of the standards marked as allowed in the corresponding column in Table 30.

Table 30 — Audiovisual packages containers referenced standards

Standard	Short designation		Extension	Interim preservation profile	Long term preservation profile	Additional constraints section
	LoC	CPP				
ISO/IEC 14496-14	"MP4_FF_2"		.mp4	allowed	allowed	
SMPTE ST 377-1	"MXF"		.mxf	allowed	allowed	
EBU TECH 3285 (Appendix A)	"WAVE"		.wav	allowed	allowed	
ISO/IEC 14496-12	"ISO_BMFF"		.heic .heif .hif .jp2 .jpf .jpm .jpx .jxl .m4a .mj2 .mjp2 .mp4	allowed	allowed	
Matroska		"MATROSKA"	.mkv	allowed	not allowed	
ISO/IEC 13818-1		"MPEG_PS"	.mpg .mpeg .m2p .ps	allowed	allowed	
ISO/IEC 13818-1		"MPEG_TS"	.ts	allowed	allowed	

9.4.5.2 Video Codecs

If the subpackage belongs to a Cinema Preservation Package with the “Long term preservation profile”, then the video codecs shall belong to one of the standards marked as allowed in the corresponding column in Table 31.

Else, for the “Interim preservation profile”, the video codecs should belong to one of the standards marked as allowed in the corresponding column in Table 31.

Table 31 — Audiovisual packages video codecs referenced standards

Standard	Short designation		Interim preservation profile	Long term preservation profile	Additional constraints section
	LoC	CPP			
ISO/IEC 14496-10	“MPEG-4_AVC”		allowed	allowed	
ISO/IEC 23008-2		“H265_HEVC”	allowed	allowed	
ISO/IEC FDIS 23090-3		“H266_VCC”	allowed	allowed	
ISO/IEC 15444-1	“J2K_C”		allowed	allowed	9.1.8.2
SMPTE RDD 36:2015		“PRORES”	allowed	not allowed	
ISO/IEC 13818-2	“MPEG-2”		allowed	allowed	
SMPTE ST 2019-1:2016		“VC3”	allowed	allowed	

9.4.5.3 Audio Codecs

If the subpackage belongs to a Cinema Preservation Package with the “Long term preservation profile”, then the audio codecs shall belong to one of the standards marked as allowed in the corresponding column in Table 32.

Else, for the “Interim preservation profile”, the audio codecs should belong to one of the standards marked as allowed in the corresponding column in Table 32.

Table 32 — Audiovisual packages audio codecs referenced standards

Standard	Short designation		Interim preservation profile	Long term preservation profile	Additional constraints section
	LoC	CPP			
	“LPCM”		allowed	allowed	
ETSI TS 102 366	“AC-3_A”		allowed	allowed	
IETF RFC 9043	“FFV1”		allowed	allowed	
ISO/IEC 13818-3	“MP3_ENC”		allowed	allowed	

9.4.6 Technical Metadata

Exactly one technical metadata file shall be present.

The technical specifics of the data files shall be documented in the metadata file as specified in 11.3.

9.5 Componentized Package

9.5.1 General description (informative)

This folder is targeted to convey componentized formats such as DCP or IMF packages.

9.5.2 General provisions

The Componentized Package folder shall follow the provisions listed in 6.4.

9.5.3 Data folder content

The Data folder shall contain only one instance of a componentized data structure.

NOTE Only one DCP can be put in a Data folder. Any supplemental DCPs shall be stored in another Componentized Package.

9.5.4 Componentized Packages standard conformance

9.5.4.1 Standard conformance

If the subpackage belongs to a Cinema Preservation Package with the “Long term preservation profile”, then each componentized package contained in a Componentized Package Data folder, shall belong to one of the standards marked as allowed in the corresponding column in Table 33.

Else, for the “Interim preservation profile”, each componentized package contained in a Componentized Package Data folder, should belong to one of the standards marked as allowed in the corresponding column in Table 33.

The file organization and content of files in the Data folder shall be compliant with the retained standard.

Table 33 — Componentized packages referenced standards

Standard	Short designation		Description	Interim preservation profile	Long term preservation profile	Additional constraints section
	LoC	CPP				
		“DCPInterop”	Interop Digital Cinema Package	allowed	not allowed	
SMPTE 429 series or ISO 26429 series		“DCPSMPTE”	Standardised definition of Digital Cinema Package	allowed	allowed	
SMPTE ST 2067-21		“IMF2E”	IMF Application #2 extended – Grand Studio Master	allowed	allowed	
SMPTE ST 2067-40		“IMF4”	IMF Application #4 – Cinema Mezzanine	allowed	allowed	
SMPTE ST 2067-50		“IMF5”	IMF Application #5 - ACES	allowed	allowed	
SMPTE RDD 45		“IMF_RDD45”	IMF Application ProRes	allowed	not allowed	

NOTE The Interop DCP format has no standardised specification.

9.5.5 Technical Metadata

Exactly one technical metadata file shall be present.

The technical specifics of the data files shall be documented in the metadata files as specified in 11.3.

9.6 Extra Package

9.6.1 General description

This folder is targeted to convey posters, unprocessed data, press kits, etc.

9.6.2 General provisions

The Extra Package folder shall follow the provisions listed in 6.4.

9.6.3 Data folder content

The file organization may be hierarchical within the data folder.

9.6.4 Extra Package filename

The name of the extra files shall follow the pattern: *<extra filename>".<extension>*.

Where:

- *<extra filename>* shall be a name chosen by the preservation facility providing the content. Only alphanumerical ASCII characters, ‘_’ and ‘-’ shall be used;
- *<extension>* is the file extension of the given format.

9.6.5 Technical Metadata

No technical metadata file shall be present.

10 Ancillary Data

10.1 General description (informative)

The Ancillary Data folder purpose is to convey information files related to the content, and is targeted for a human reader.

10.2 Root Ancillary Data

10.2.1 General description (informative)

The Root Ancillary Data folder purpose is to convey information files related to the whole package and enclosed work and targeted for a human reader.

10.2.2 File conformance

If the Cinema Preservation Package profile is the “Long term preservation profile”, then the Root Ancillary Data files shall belong to one of the standards marked as allowed in the corresponding column, as listed in Table 34.

Else, for the “Interim preservation profile”, the Root Ancillary Data files should belong to one of the standards marked as allowed in the corresponding column in Table 34.

Table 34 — Ancillary data recommended standards

Standards	Short designation		Interim preservation profile	Long term preservation profile	Additional constraints section
	LoC	CPP			
ISO 19005-1	"PDF/A-1"		allowed	allowed	
ISO 19005-2	"PDF/A-2"		allowed	allowed	
ISO 19005-4		"PDF/A-4"	allowed	allowed	
ISO 12234-2	"TIFF_6"		allowed	allowed	
IEEE 1003.1 ISO/IEC 10646		"TXT"	allowed	allowed	10.2.3

10.2.3 Constraints on text files

The text files shall contain only UTF-8 characters and the 127 first characters shall be limited with the constraints listed in Posix specifications for text files. No Byte order mark (BOM) shall be used. The newline character shall be "LF".

10.3 Subpackage Ancillary Data

10.3.1 General description (informative)

The Subpackage Ancillary Data folder purpose is to convey information files related to a specific subpackage and targeted for a human reader.

10.3.2 File conformance

If the subpackage belongs to a Cinema Preservation Package with the "Long term preservation profile", then the Subpackage Ancillary Data files shall belong to one of the standards marked as allowed in the corresponding column, as listed in Table 34.

Else, for the "Interim preservation profile", the Subpackage Ancillary Data files should belong to one of the standards marked as allowed in the corresponding column in Table 34.

11 Metadata

11.1 General metadata files description

11.1.1 General description (informative)

Metadata files are present in the Root Metadata folder and Subpackage Metadata folders.

11.1.2 Languages

For language descriptions the codes of the ISO 639-3 standard shall be used.

NOTE The 'xml:lang' attribute is meant to be filled with IANA language codes as provisioned by IETF RFC 3066, itself based on ISO 639-* codes.

language definition namespace shall be "ISO 639-3:2007".

language definition link shall be "<https://www.iso.org/standard/39534.html>".

11.1.3 Character encoding

The string 'UTF-8' shall be used to describe UTF-8 text file content.

11.2 Root Metadata

11.2.1 Descriptive metadata

11.2.1.1 File conformance

11.2.1.1.1 General

The root descriptive metadata files shall conform to the XML standard and follow the EBUCore XML Schema.

11.2.1.1.2 First Root Descriptive Metadata file content

The first root descriptive metadata file shall enclose the work descriptive metadata as specified in EN 15744. Following the naming specification in 6.5.4, the name of this file is:

"descMD_<uuid>_0000.xml", where <uuid> is the Preservation Package UUID.

11.2.1.1.3 Other Root Descriptive Metadata files content

Other variant, manifestation or other kind of version descriptive metadata files, using the EBUCore XML schema, are allowed and shall follow the constraints on variants descriptive metadata.

11.2.1.2 Constraints on the Work Descriptive Metadata file

11.2.1.2.1 Constraints on the root `ebuCoreMain` element

XPath is: `"/ebucore:ebuCoreMain"`.

The `<ebucore:ebuCoreMain>` root element shall include the following namespaces:

- `"urn:ebu:metadata-schema:ebucore"`;
- `"https://purl.org/dc/elements/1.1/"`.

The following prose in this section assume that the used prefix for these namespaces are respectively "ebucore", and "dc", elements without namespaces belong to the EBUCore schema.

The elements listed in Table 35 shall be present with the specified arity.

Table 35 — <ebuCoreMain> children for work descriptive metadata

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
			@dateLastModified	x	<i>last modification date</i>
			@timeLastModified	x	<i>last modification time</i>
	coreMetadata	1	#children	x	See 11.2.1.2.2
Source	metadataProvider	1	#children	x	organisationDetails
	metadataProvider/organisationDetails	1	<i>source organization details element</i>		

last modification date shall be the date the content of the file has last been modified.

last modification time shall be the time of the day the content of the file has last been modified.

The source element documents the facility providing the metadata; *source organization details element* shall use the organisationDetailsType from EBUCore to describe this organization.

11.2.1.2.2 Constraints on the <coreMetadata> element

XPath is: “/ebuCore:ebuCoreMain/ebuCore:coreMetadata”.

The elements listed in Table 36 shall be present with the specified arity.

Table 36 — <coreMetadata> children for work descriptive metadata

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Title	title	1	@typeLabel	x	<i>title type</i>
			#children	x	dc:title
	title/dc:title	1	@xml:lang	o	<i>title name language</i>
			#text	x	<i>title name</i>
Alternative title	alternativeTitle	0..n	@typeLabel	x	<i>alternative title type</i>
			#children	x	dc:title
	alternativeTitle/dc:title	1	@xml:lang	o	<i>alternative title name language</i>
			#text	x	<i>alternative title name</i>
Cast	contributor	0..n	#children	o	contactDetails, role
	contributor/contactDetails	1..n	@contactId	x	<i>cast contactId</i>
			#children	x	name
	contributor/contactDetails/name	1	#text	x	<i>cast contributor name</i>
Credits	contributor/role	1	@typeLabel	o	“cast”
	contributor	0..n	#children	o	contactDetails, role
			#children	o	
	contributor/contactDetails	1..n	@contactId	x	<i>credit contactId</i>
			#children	x	name
	contributor/contactDetails/name	1	#text	x	<i>credit contributor name</i>
Production company	contributor/role	1	@typeLabel	o	“credits”
	contributor/role	1	@typeLabel	o	<i>credit role</i>
	contributor/role	1	@typeLabel	o	<i>credit role</i>
Production company	publisher	0..n	#children	x	organisationDetails, role
	publisher/organisationDetails	1..n	@organisationId	x	<i>production organizationId</i>

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
			#children	x	organizationName
	publisher/organisationDetails/organizationName	1	#text	x	<i>production organization name</i>
	publisher/role	1	@typeLabel	x	"productionCompany"
Country of reference	coverage	0..n	#children	x	spatial
	coverage/spatial	1	#children	x	location
	coverage/spatial/location	1	@typeLabel	x	"countryOfReference"
			@locationId	o	<i>reference country locationId</i>
			#children	x	name
	coverage/spatial/location/name	1	@xml:lang	o	<i>location name language</i>
			#text	x	<i>location name</i>
Language	language	0..n	@typeLabel	x	"originalLanguage"
			#children	x	dc:language
	language/dc:language	1	#text	x	<i>original language</i>
Year of reference	date	1..n	@typeLabel	x	"yearOfReference"
			#children	x	created
	date/created	1	@startYear	x	<i>start year of reference</i>
			@endYear	x	<i>end year of reference</i>
Identifier	identifier	1..n	@typeLabel	o	<i>identifierKind</i>
			@formatLabel	x	"URI"
			#children	x	dc:identifier
	identifier/dc:identifier	1	#text	x	<i>identifier</i>
Genre	type	0..1	#children	x	genre
	type/genre	1	@typeLabel	x	<i>genre</i>
Relationship	isRelatedTo	0..n	#children	x	relationIdentifier
	isRelatedTo/relationIdentifier	1	#children	x	dc:identifier
	isRelatedTo/relationIdentifier/dc:identifier	1	#text	x	<i>relation identifier</i>

title name shall be the main title of the work within the CPP Package, the language used can be specified using title name language. The title type should be one of the following value: "workingTitle", "originalTitle", "releaseTitle", "archiveTitle".

The work main title can be completed with one or more alternate title. For each alternate title, alternative title name shall be another alternate title, the language used can be specified using alternative title name language. The alternative title type should be one of the following value: “workingTitle”, “originalTitle”, “releaseTitle”, “archiveTitle”, “series”.

Multiple cast contributors can be added, using cast contributor name. Cast contactId shall be filled with an identifier of the contact. The contactId should use a common and permanent identification vocabulary.

Multiple credit contributors can be added, using credit contributor name. Credit contactId shall be filled with an identifier of the contact. The contactId should use a common and permanent identification vocabulary. The credit role shall contain a description of the credit role.

EXAMPLE 1 possible *credit role* values are “director”, “screenplay”, “editor”, “directorOfPhotography”, etc.

One or more publisher with “productionCompany” typeLabel attribute may be added. The *production organization name* shall be the name of the production company. The *production organizationId* shall be filled with an identifier of the organization. The *organizationId* should use a common and permanent identification vocabulary.

EXAMPLE 2 ISNI identifiers are possible persons and institutions identifiers.

reference country locationId shall be an identifier to a country. The *locationId* shall be an ISO 3166-1 alpha-2 code, or a four-letter ISO 3166-3 code for countries that no longer exist. The *location name* shall be the official name of the country in *location name language*, accordingly.

original language shall be the main language code.

start year of reference and *end year of reference* shall be the start year and end year of reference for the work. Both values may be the same.

At least one work identifier shall be present. Each *identifier* shall be an URI. The first identifier shall not enclose the typeLabel attribute, the other ones shall enclose the typeLabel attribute, the *identifierKind* shall be “alternative” in this case.

genre shall be a value denoting the genre of the movie content.

EXAMPLE 3 “Fiction” is a possible value.

For each <relationIdentifier> element, *relation identifier* shall be an identifier related to a related object. The same identification schema shall be used as for the identifier element.

11.2.1.3 Constraints on the variant descriptive metadata files

11.2.1.3.1 Constraints on the root <ebuCoreMain> element

XPath is: “/ebuCoreMain”.

The elements listed in Table 37 shall be present with the specified arity.

Table 37 — ebuCoreMain children for variant descriptive metadata

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
			@dateLastModified	x	<i>last modification date</i>
			@timeLastModified	x	<i>last modification time</i>
	coreMetadata	1	#children	x	See 11.2.1.3.2
source	metadataProvider	1	#children	x	<i>organisationDetails</i>
	metadataProvider/ organisationDetails	1	<i>source organization details element</i>		

last modification date shall be the date the content of the file has last been modified.

last modification time shall be the time of the day the content of the file has last been modified.

The source element documents the facility providing the metadata; source organization details element shall use the *organisationDetailsType* from EBUCore to describe this organization.

11.2.1.3.2 Constraints on the <coreMetadata> element

XPath is: “/ebucore:ebuCoreMain/ebucore:coreMetadata”.

The elements listed in Table 38 shall be present with the specified arity.

Table 38 — coreMetadata children for variant descriptive metadata

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Title	title	1	@typeLabel	x	<i>title type</i>
			#children	x	dc:title
	title/dc:title	1	@xml:lang	o	<i>title name language</i>
			#text	x	<i>title name</i>
Identifier	identifier	1..n	@typeLabel	o	<i>identifierKind</i>
			@formatLabel	x	“URI”
			#children	x	dc:identifier
	identifier/dc:identifier	1	#text	x	<i>identifier</i>
Version of	isVersionOf	1..n	#children	x	relationIdentifier
	isVersionOf/relationIdentifier	1	#children	x	dc:identifier
	isVersionOf/relationIdentifier/ dc:identifier	1	#text	x	<i>version relation identifier</i>
Has data object	hasPart	1..n	#children	x	relationIdentifier
	hasPart/relationIdentifier	1	#children	x	dc:identifier
	hasPart/relationIdentifier/ dc:identifier	1	#text	x	<i>data object relation identifier</i>
Same as	isRelatedTo	0..n	#children	x	relationIdentifier

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
	isRelatedTo/relationIdentifier	1	@typeLabel	x	"sameAs"
			#children	x	dc:identifier
	isRelatedTo/relationIdentifier/dc:identifier	1	#text	x	<i>same object relation identifier</i>
Release date	date	0..1	#children	x	released
	date/released	1	@year	x	<i>release year</i>
Production year	date	0..1	#children	x	created
	date/created	1	@startYear	x	<i>production start year</i>
			@endYear	x	<i>production end year</i>
Manifestation type	type	0..1	#children	x	objectType
	type/objectType	1	@typeLabel	x	<i>objectTypeType</i>
HasAgent (corporate body)	contributor	0..n	#children	x	organisationDetails, role
	contributor/organisationDetails	1	@organisationId	x	<i>agent organisationId</i>
			#children	x	organisationName
	contributor/organisationDetails/organisationName	1	#text	x	<i>agent organization name</i>
	contributor/role	1	@typeLabel	x	"source"

title name shall be the main title of the version the descriptive metadata are related to, the language used can be specified using title name language. The title type should be one of the following value: "workingTitle", "originalTitle", "releaseTitle", "archiveTitle".

At least one version identifier shall be present. Each identifier shall be an URI.

identifierKind shall be a string describing the kind of identifier used, either local to the facility or external.

version relation identifier shall point to the work identifier as specified in 6.9.

For each <hasPart> element, data object relation identifier shall be an identifier related to a related object. The same identification schema shall be used as for the identifier element.

For each <isRelatedTo> element with "sameAs" typeLabel attribute, same object relation identifier shall be an identifier related to a related object. The same identification schema shall be used as for the identifier element.

release year shall be the release year of the version.

production start year and production end year shall be respectively the start year and end year of production for the version. Both values may be the same.

objectTypeType shall describe the relationship between the manifestation and the cinematographic work (or a variant).

One or more agent may be added as metadata provider. The agent organization name shall be the name of the facility. The agent organisationId shall be filled with an identifier of the organization. The organizationId should use a common and permanent identification vocabulary.

11.2.2 Provenance metadata

The provenance metadata files shall conform to the XML standard and follow the PREMIS XML Schema.

11.3 Subpackage metadata

11.3.1 Technical metadata

11.3.1.1 General file conformance

The technical metadata files shall conform to the XML standard and follow the EBUCore XML Schema with additional constraints as specified in Table 39.

The subpackage technical metadata files shall be encoded using UTF-8 character encoding.

Table 39 — Applicable constraints applicable to schema per subpackage type

Subpackage type	Additional constraints section
Image Package	See 11.3.1.2 and 11.3.1.3
Sound Package	See 11.3.1.2 and 11.3.1.4
Timed Text Package	See 11.3.1.2 and 11.3.1.5
Audiovisual Package	See 11.3.1.2 and 11.3.1.6
Componentized Package	See 11.3.1.2 and 11.3.1.7

11.3.1.2 General constraints

11.3.1.2.1 Constraints on the <ebuCoreMain> element

The <ebuCore:ebuCoreMain> root element shall include the following namespaces:

- “urn:ebu:metadata-schema:ebuCore”;
- “<https://purl.org/dc/elements/1.1/>”.

The following prose in this section assume that the used prefix for these namespaces are respectively “ebuCore”, and “dc”, elements without namespaces belongs to the EBUCore schema.

11.3.1.2.2 Constraints on the <coreMetadata> element

XPath is: “/ebuCore:ebuCoreMain/ebuCore:coreMetadata”.

The elements listed in Table 40 shall be present with the indicated arity.

Table 40 — <coreMetadata> optional children

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Subpackage name	title	0..1	@typeLabel	x	"subpackageName"
			#children	x	dc:title
	title/dc:title		#text	x	<i>subpackage name</i>
File origin	description	0..1	@typeLabel	x	"fileOrigin"
			@typeNamespace	x	"en17650:2022"
			@typeLink	x	<i>fileOriginTypeLink</i>
			#children	x	dc:description
	description/dc:description	1	#text	x	<i>fileOriginContent</i>
Annotation text	description	0..1	@typeLabel	x	"userText"
			#children	x	dc:description
	description/dc:description	1	#text	x	<i>annotation text</i>

subPackageName, fileOriginTypeLink and fileOriginContent are specified in the following specific metadata sections.

annotation text shall be a short description of the content.

Additional elements specified in the following specific metadata sections shall be present.

11.3.1.2.3 Format name specification

The file format names should be either those listed in the format dictionary of the Library of Congress, or in the PRONOM technical registry. The name of the format should be the "short name" in the dictionary format entry or the PRONOM Unique Identifier (PUID) respectively.

In any case the format names shall be listed in a controlled vocabulary.

NOTE 1 The formats listed by the Library of congress can be found at <https://www.loc.gov/preservation/digital/formats/fdd/>

NOTE 2 The formats listed by the PRONOM UK Archive technical registry can be found at <https://www.nationalarchives.gov.uk/PRONOM/>

11.3.1.2.4 File origin default enumeration link and values

The following sections are defining file origin values for specific subpackages.

fileOriginTypeLinkDefaultValue shall be "urn:cen.eu:en17650:2022:ns:metadata#commonFileOrigin".

fileOriginContentDefaultValueList shall be a list with the following values: "MechanicalAnalog", "MechanicalDigital", "MagneticAnalog", "MagneticDigital", "OpticalAnalog", "OpticalDigital", "FileBased", "MixedOrigin", "Unknown".

11.3.1.3 Specific constraints for Image Package

11.3.1.3.1 Constraints on the coreMetadata element

XPath is: "/ebuCore:ebuCoreMain/ebuCore:coreMetadata".

The elements listed in 11.3.1.2 shall be present.

subPackageName shall be the name of the subpackage.

fileOriginTypeLink should be the *fileOriginTypeLinkDefaultValue*, as specified in 11.3.1.2.4.

If the *fileOriginTypeLink* is set to the recommended URI, *fileOriginContent* shall be one of the values included in *fileOriginContentDefaultValueList*, as specified in 11.3.1.2.4.

EXAMPLE For a film origin, a possible value is “OpticalAnalog”.

The elements listed in Table 41 shall be present.

Table 41 — <coreMetadata> mandatory children for Image Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	Value(s)
Image Package Format		1	@formatName	x	“imagePackageFormat”
			#children	x	see 11.3.1.3.2
Container format	format	1	@formatName	x	“containerFormat”
			#children	x	see 11.3.1.3.3
Image format	format	1	@formatName	x	“imageFormat”
			#children	x	see 11.3.1.3.5

11.3.1.3.2 Constraints on the “imagePackageFormat” <format> element

XPath is:

“/ebuCore:ebuCoreMain/ebuCore:coreMetadata/ebuCore:format[@formatName='imagePackageFormat']”.

Exactly one of this element shall be present.

The elements listed in Table 42 shall be present.

Table 42 — “imagePackageFormat” <format> mandatory children for Image Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Duration	duration	1	@typeLabel	x	“durationInFrames”
			#children	x	editUnitNumber
	duration/editUnitNumber	1	#text	x	edit unit number
Framerate	editRate	1	@factorNumerator	o	edit rate numerator
			@factorDenominator	o	edit rate denominator
			#text	x	integer edit rate
Filename Prefix	technicalAttributeString	1	@typeLabel	x	“filenamePrefix”
			#text	x	prefix of image sequence filenames

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Filename Extension	technicalAttributeString	1	@typeLabel	x	"filenameExtension"
			#text	x	<i>extension of image sequence filenames</i>
Filename Digits	technicalAttributeUnsignedByte	1	@typeLabel	x	"filenameDigits"
			#text	x	<i>number of digits in sequence filenames</i>
First Index	technicalAttributeUnsignedInteger	1	@typeLabel	x	"firstIndex"
			#text	x	<i>first file index</i>
Last Index	technicalAttributeUnsignedInteger	1	@typeLabel	x	"lastIndex"
			#text	x	<i>last file index</i>
First useful frame	technicalAttributeUnsignedInteger	0..1	@typeLabel	x	"firstActiveIndex"
			#text	x	<i>first active file index</i>
Last useful frame	technicalAttributeUnsignedInteger	0..1	@typeLabel	x	"lastActiveIndex"
			#text	x	<i>last active file index</i>

edit unit number shall be the duration in number of edit units.

NOTE 1 An edit unit can be composed of one or more images.

EXAMPLE 1 interlaced stereoscopic sequences use two image files per edit unit.

The <editRate> element shall have a child text with an *integer edit rate* value. In case of fractional edit rates, *edit rate numerator* and *edit rate denominator* shall both be present.

EXAMPLE 2 For 29.970 fps, the *integer edit rate* child text is 30, *edit rate numerator* is 1000 and *edit rate denominator* is 1001.

prefix of image sequence filenames shall be *image prefix* as specified in 9.1.6.

extension of image sequence filenames shall be *extension* as specified in 9.1.6.

number of digits in sequence filenames should be 8.

first file index shall be the first file number used in the Data folder.

last file index shall be the last file number used in the Data folder.

first active file index shall be an integer value representing the number of the first frame of active content.

last active file index shall be an integer value representing the number of the last frame of active content.

NOTE 2 active content is the visible and audible part of a content.

11.3.1.3.3 Constraints on the "containerFormat" <format> element

XPath is:

"/ebucore:ebuCoreMain/ebucore:coreMetadata/ebucore:format[@formatName='containerFormat']".

Exactly one of this element shall be present.

The elements listed in Table 43 shall be present.

Table 43 — “containerFormat” <format> children for Image Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	Value(s)
Container format metadata	containerFormat	1	@containerFormatName	x	<i>formatName</i>
			@containerFormatDefinition	o	<i>formatVersionName</i>
			#children	x	See 11.3.1.3.4
MIME type	mimeType	0..1	@typeLabel	x	<i>mime type</i>

formatName shall be as specified in 11.3.1.2.3.

formatVersionName shall be a string denoting the version.

EXAMPLE “2.0”, “2018”.

mime type shall be the common MIME type of all images.

11.3.1.3.4 Constraints on the <containerFormat> element

XPath is:

“/ebuCore:ebuCoreMain/ebuCore:coreMetadata/ebuCore:format[@formatName='containerFormat']/ebuCore:containerFormat”.

The elements listed in Table 44 shall be present.

Table 44 — children of the <containerFormat> element

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Container standard reference	technicalAttributeString	1	@typeLabel	x	“containerStandardReference”
			#text	x	<i>description of container standard</i>
Container width	technicalAttributeUnsignedInteger	1	@typeLabel	x	“containerWidth”
			#text	x	<i>containerWidth</i>
Container height	technicalAttributeUnsignedInteger	1	@typeLabel	x	“containerHeight”
			#text	x	<i>containerHeight</i>
Container standard reference	technicalAttributeUri	1	@typeLabel	x	“containerStandardReference”
			#text	x	<i>link to container standard reference</i>

description of container standard shall be the name of the container standard. If the standard cannot be identified, then it shall be the string “Unknown”.

EXAMPLE 1 SMPTE ST 268:2014 - SMPTE Standard - File Format for Digital Moving-Picture Exchange.

containerWidth shall be the width of the image container. *containerHeight* shall be the height of the image container. These are not necessarily the dimensions of the active area.

link to container standard reference shall be a permanent link to the standard. If the standard cannot be identified, then it shall be the following URI:

“urn:cen.eu:en17650:2022:ns:metadata#UnknownStandard”.

EXAMPLE 2 <https://doi.org/10.5594/SMPTE.ST268.2014>.

11.3.1.3.5 Constraints on the “imageFormat” <format> element

XPath is:

“/ebuCore:ebuCoreMain/ebuCore:coreMetadata/ebuCore:format[@formatName='imageFormat']”.

The element listed in Table 45 shall be present.

Table 45 — “imageFormat” <format> children for Image Packages

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	Value(s)
Image format metadata	imageFormat	1	#children	x	See 0

11.3.1.3.6 Constraints on the <imageFormat> element

XPath is:

“/ebuCore:ebuCoreMain/ebuCore:coreMetadata/ebuCore:format[@formatName='imageFormat']/ebuCore:imageFormat”.

The elements listed in Table 46 shall be present.

Table 46 — <imageFormat> children for Image Packages

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Active region	regionDelimX	0..1	#text	x	<i>bottom left corner of the region</i>
	regionDelimY	0..1	#text	x	
	width	0..1	#text	x	<i>active width</i>
	height	0..1	#text	x	<i>active height</i>
Aspect ratio	aspectratio	0..1	#children	x	factorNumerator, factorDenominator
	aspectratio/factorNumerator		#text	x	<i>ratioNumerator</i>
	aspectratio/factorDenominator		#text	x	<i>ratioDenominator</i>
Image codec standard reference	technicalAttributeString	1	@typeLabel	x	“imageCodecStandardReference”
			#text	x	<i>codec standard description</i>
Colour standard reference	technicalAttributeString	1	@typeLabel	x	“colourReference”
			#text	x	<i>colour standard description</i>

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Chroma Subsampling	technicalAttributeUnsignedByte	0..1	@typeLabel	x	"chromaSubsampling"
			#text	x	<i>chromaSubsampling</i>
Progressive Mode	technicalAttributeUnsignedByte	0..1	@typeLabel	x	"progressiveMode"
			#text	x	<i>progressiveMode</i>
Number of colour components	technicalAttributeUnsignedInteger	1	@typeLabel	x	"componentNumber"
			#text	x	<i>Number of colour components</i>
Bitdepth of colour components	technicalAttributeUnsignedInteger	1	@typeLabel	x	"componentBitdepth"
			#text	x	<i>bitdepth of components</i>
Number of channels	technicalAttributeUnsignedInteger	1	@typeLabel	x	"channelNumber"
			#text	x	<i>channelCount</i>
Index Colour Primaries	technicalAttributeUnsignedInteger	1	@typeLabel	x	"colourPrimaries"
			#text	x	<i>Index of colour primaries according to ITU-T H.273</i>
Index Transfer Characteristics	technicalAttributeUnsignedInteger	1	@typeLabel	x	"transferCharacteristics"
			#text	x	<i>Index of transfer characteristics according to ITU-T H.273</i>
Index Matrix Coefficients	technicalAttributeUnsignedInteger	1	@typeLabel	x	"matrixCoefficients"
			#text	x	<i>Index of matrix coefficients according to ITU-T H.273</i>
Video Full Range flag	technicalAttributeUnsignedInteger	1	@typeLabel	x	"fullRangeFlag"
			#text	x	<i>full range flag</i>
Luminance Min	technicalAttributeUnsignedInteger	0..1	@typeLabel	x	"luminanceMin"
			#text	x	<i>mimimum Luminance of display</i>
Luminance Max	technicalAttributeUnsignedInteger	0..1	@typeLabel	x	"luminanceMax"
			#text	x	<i>maximum Luminance of display</i>
Value Range Min	technicalAttributeUnsignedInteger	0..1	@typeLabel	x	"valueRangeMin"
			#text	x	<i>mimimum Value (black value)</i>
Value Range Max	technicalAttributeUnsignedInteger	0..1	@typeLabel	x	"valueRangeMax"
			#text	x	<i>maximum Value (white value)</i>
Colour standard reference	technicalAttributeUri	1	@typeLabel	x	"colourReference"
			#text	x	<i>colourReferenceLink</i>
Image codec standard reference	technicalAttributeUri	1	@typeLabel	x	"imageCodecStandardReference"
			#text	x	<i>codec standard link</i>

If the image width and height are smaller than the values containerWidth and containerHeight in the container section 11.3.1.3.4, then the active area shall be indicated with delimX and delimY, indicating the bottom left corner of the region, and width and height indicating the active width and height of the image. All width, height, delimX, delimY shall be present or none.

NOTE 1 the active area is the visible part of an image.

The active image ratio is documented using integer rational parameters *ratioNumerator* and *ratioDenominator*.

EXAMPLE 1 A Cinemascope ratio is indicated by *ratioNumerator* = "239", *ratioDenominator* = "100". An HD ratio is indicated by *ratioNumerator* = "16", *ratioDenominator* = "9".

NOTE 2 In case of a non-square pixel aspect ratio, it can be deduced from the active area size and the image ratio.

codec standard description shall be the name of the codec standard. If the standard cannot be identified, then it shall be the string "Unknown".

colour standard description shall be a string describing the colour standard. If the standard cannot be identified, then it shall be the string "Unknown".

EXAMPLE 2 "ITU-R BT.601"

chromaSubsampling shall be one of the value in D.2.

progressiveMode shall be one of the value in D.1.

number of colour components shall be the number of colour primaries used in the colour representation.

bitdepth of components shall be the bit depth of components. If some components are subsampled, the larger component bit depth shall be used.

channelCount shall be the number of channels. Alpha channels, depth map channels or other additional channels shall be included in this number if present.

The "colourPrimaries", "transferCharacteristics", "matrixCoefficients", "fullRangeFlag" technicalAttributeUnsignedInteger elements are mandatory for trichromatic images.

Index of colour primaries according to ITU-T H.273 shall be the index of the ITU-T H.273:2016, Table 2 standard.

NOTE 3 If the colour primaries are unknown or determined otherwise, the value is 2 for Unspecified.

Index of transfer characteristics according to ITU-T H.273 shall be the index of the ITU-T H.273:2016, Table 3 standard.

NOTE 4 If the transfer characteristics are unknown or determined otherwise, the value shall be 2 for Unspecified.

Index of matrix coefficients according to ITU-T H.273 shall be the index of the ITU-T H.273:2016, Table 4 standard.

NOTE 5 If the matrix coefficients are unknown or determined otherwise, the value shall be 2 for Unspecified.

full range flag shall be "1" if the image components values are full range, and "0" else. In the latter case, the <valueRangeMin> and <valueRangeMax> elements shall be present, and the *minimum range value* shall be the minimum possible image pixel value, and *maximum range value* shall be the maximum possible pixel value.

minimum Luminance of display shall be the minimum luminance of the reference display.

maximum Luminance of display shall be the maximum luminance of the reference display.

The intended display luminance range shall be specified for HDR content.

colourReferenceLink shall be a permanent link to the colour standard. If the standard cannot be identified, then it shall be the following URI: "urn:cen.eu:en17650:2022:ns:metadata#UnknownStandard".

EXAMPLE 3 “https://www.itu.int/rec/R-REC-BT.601-7-201103-I/en”.

codec standard link shall be a permanent link to the standard. If the standard cannot be identified, then it shall be the following URI: “urn:cen.eu:en17650:2022:ns:metadata#UnknownStandard”.

NOTE 6 If the codec is not specified in a separate standard, it can be the same standard as the container.

11.3.1.4 Specific constraints for Sound Package

11.3.1.4.1 Constraints on the <coreMetadata> element

XPath is: “/ebuCore:ebuCoreMain/ebuCore:coreMetadata”.

The elements listed in 11.3.1.2.2 shall be present.

subPackageName shall be the name of the subpackage.

fileOriginTypeLink should be the *fileOriginTypeLinkDefaultValue*, as specified in 11.3.1.2.4.

If the *fileOriginTypeLink* is set to the recommended URI, *fileOriginContent* shall be one of the values included in *fileOriginContentDefaultValueList*, as specified in 11.3.1.2.4.

The elements listed in Table 47 shall be present.

Table 47 — <coreMetadata> mandatory children for Sound Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	Value(s)
Sound package format	format	1	@formatName	x	“soundPackageFormat”
			#children	x	see 11.3.1.4.2
Container format	format	1	@formatName	x	“containerFormat”
			#children	x	see 11.3.1.4.3
Audio format	format	1	@formatName	x	“audioFormat”
			#children	x	see 11.3.1.4.5

11.3.1.4.2 Constraints on the “soundPackageFormat” <format> element

XPath is:

“/ebuCore:ebuCoreMain/ebuCore:coreMetadata/ebuCore:format[@formatName='soundPackageFormat']”.

This element shall be present and shall be unique.

The elements listed in Table 48 shall be present.

Table 48 — “soundPackageFormat” <format> children for Sound Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Duration	duration	0..1	#children	x	normalPlayTime
	duration/normalPlayTime	1	#text	x	<i>playTime</i>
	duration	0..1	#children	x	editUnitNumber
	duration/editUnitNumber	1	editRate	o	<i>edit rate</i>

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
			factorNumerator	o	<i>correction numerator</i>
			factorDenominator		<i>correction denominator</i>
			#text	x	<i>durationInEditNumber</i>
Filename Prefix	technicalAttributeString	1	@typeLabel	x	"filenamePrefix"
			#text	x	<i>prefix of sound filenames</i>
Filename Extension	technicalAttributeString	1	@typeLabel	x	"filenameExtension"
			#text	x	<i>extension of sound filenames</i>
Filename Digits	technicalAttributeUnsignedByte	1	@typeLabel	x	"filenameDigits"
			#text	x	<i>number of digits in sound filenames</i>
First active sample	technicalAttributeUnsignedInteger	0..1	@typeLabel	x	"firstActiveSample"
			#text	x	<i>firstActiveSample</i>
Last active sample	technicalAttributeUnsignedInteger	0..1	@typeLabel	x	"lastActiveSample"
			#text	x	<i>lastActiveSample</i>

playTime shall be the play time of the content expressed in the form HH:MM:SS.S.

As an option, the framerate of the related image content may be indicated.

If present *edit rate* shall be the edit rate in frames per second without correction.

If present *correction numerator* shall be the correction numerator and *correction denominator* shall be the correction denominator.

durationInEditNumber shall be the number of edit units as specified in EBUCore.

NOTE According to EBUCore documentation, the editUnitNumber is a complex type to express a number of edit units. An editUnit is the inverse of the edit rate, or corrected edit rate as the result of:

$$\text{editUnit} = \frac{1}{\text{edit rate} \times \frac{\text{correction numerator}}{\text{correction denominator}}}$$

EXAMPLE For a fractional frame rate of 29.97fps the editRate = "30", correction numerator = "1000", and correction denominator = "1001".

prefix of sound filenames shall be the *sound file prefix* as specified in 9.2.7.

extension of sound filenames is the *extension* as specified in 9.2.7.

number of digits in sound filenames shall be 2.

firstActiveSample is the first sample part of the active content.

lastActiveSample is the last sample part of the active content.

11.3.1.4.3 Constraint on the “containerFormat” format element

XPath is:
 “/ebuCore:ebuCoreMain/ebuCore:coreMetadata/ebuCore:format[@formatName='containerFormat']”.

This element shall be present and shall be unique.

The elements listed in Table 49 shall be present.

Table 49 — “containerFormat” format children for Sound Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	Value(s)
Container format metadata	containerFormat	1	@containerFormatName	x	<i>formatName</i>
			@containerFormatDefinition	o	<i>formatVersionName</i>
			#children	x	See 11.3.1.4.4
Mime type	mimeType	0..1	@typeLabel	x	<i>mime type</i>

formatName shall be the format as specified in 11.3.1.2.3.

formatVersionName shall be a string denoting the version.

EXAMPLE “2.0”, “2018”.

mime type shall be the common MIME type of all sound files in the subpackage.

11.3.1.4.4 Constraint on the <containerFormat> element

XPath is:

“/ebuCore:ebuCoreMain/ebuCore:coreMetadata/ebuCore:format[@formatName='containerFormat']/ebuCore:containerFormat”.

The elements listed in Table 50 shall be present.

Table 50 — <containerFormat> children for Sound Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Container standard reference	technicalAttributeString	1	@typeLabel	x	“containerStandardReference”
			#text	x	<i>description of container standard</i>
Container standard reference	technicalAttributeUri	1	@typeLabel	x	“containerStandardReference”
			#text	x	<i>link to container standard reference</i>

description of container standard shall be the name of the container standard. If the standard cannot be identified, then it shall be the string “Unknown”.

EXAMPLE 1 BS.1352: File format for the exchange of audio programme materials with metadata on information technology media.

link to container standard reference shall be a permanent link to the standard. If the standard cannot be identified, then it shall be the following URI:

“urn:cen.eu:en17650:2022:ns:metadata#UnknownStandard”.

EXAMPLE 2 <https://www.itu.int/rec/R-REC-BS.1352-3-200712-I/en>.

11.3.1.4.5 Constraint on the “audioFormat” <format> element

XPath is:

“/ebucore:ebuCoreMain/ebucore:coreMetadata/ebucore:format[@formatName='audioFormat']”.

The element listed in Table 51 shall be present.

Table 51 — “audioFormat” format children for Sound Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	Value(s)
Audio format metadata	audioFormat	1	@audioFormatName	x	<i>audio format name</i>
			@audioFormatDefinition	o	<i>audio format version</i>
			#children	x	See 11.3.1.4.6

audio format name shall be the format name as specified in 11.3.1.2.3.

audio format version shall be a string denoting the version.

EXAMPLE “2.0”, “2018”.

11.3.1.4.6 Constraint on the <audioFormat> element

XPath is:

“/ebucore:ebuCoreMain/ebucore:coreMetadata/ebucore:format[@formatName='audioFormat']/ebucore:audioFormat”.

The elements listed in Table 52 shall be present.

Table 52 — audioFormat children for Sound Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Audiotrack configuration	audioTrackConfiguration	1	@typeNamespace	x	“en17650:2022”
			@typeLink	x	“urn:cen.eu:en17650:2022:ns:metadata#audioTrackConfiguration”
			@typeLabel	x	<i>audioTrackConfiguration</i>
Sample rate	samplingRate	1	#text	x	<i>sampling frequency</i>
Sample Bit Depth	sampleSize	1	#text	x	<i>sample size</i>
Sample Type	sampleType	1	#text	x	<i>ebuSampleType</i>
Track description	audioTrack	1	@trackLanguage	o	<i>trackLanguage</i>
			@trackId	x	<i>trackId</i>
			@trackName	x	<i>trackName</i>
Number of channels	channels	1	#text	x	<i>channelsCount</i>

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Audio Codec standard reference	technicalAttribute String	1	@typeLabel	x	"audioCodecStandardReference"
			#text	x	<i>description of codec standard</i>
Audio Track kind	technicalAttribute String	1	@typeLabel	x	"audioContentDescription"
			@typeNamespace	x	"en17650:2022"
			@typeLink	x	"urn:cen.eu:en17650:2022:ns:metadata#soundFileTrackKind"
			#text	x	<i>audioTrackKind</i>
Channel layout	technicalAttribute String	1	@typeLabel	x	"channelLayout"
			#text	x	<i>channelLayout</i>
Channel location	technicalAttribute String	1	@typeLabel	x	"channelLocation"
			#text	x	<i>channelLocation</i>
Loudness	technicalAttribute String	0..1	@typeLabel	x	"loudnessMethod"
			@typeNamespace	x	"en17650:2022"
			@typeLink	x	"urn:cen.eu:en17650:2022:ns:metadata#soundLoudness"
			#text	x	<i>integratedLoudnessMethod</i>
	technicalAttribute Float	0..1	@typeLabel	x	"integratedLoudness"
			#text	x	<i>integratedLoudnessValue</i>
Audio Codec standard reference	technicalAttribute Uri	1	@typeLabel	x	"audioCodecStandardReference"
			#text	x	<i>link to codec standard reference</i>

audioTrackConfiguration shall be <X>.<Y>["+"<Z>], where:

- <X> is the number of discrete channels;
- <Y> is the number of additional number of subwoofer channels;
- <Z> is the number of additional channels.

EXAMPLE 1 "2.0", "5.1+1".

sampling frequency shall be the sample rate.

sample size shall be the bit depth of the samples.

ebuSampleType shall be "float" or "integer" as provisioned by the EBUCore specification.

trackLanguage shall be a language code.

trackId specifies the number of the track in order of the appearance in the file, or a track label if available in the file.

trackName shall be a user specified value characterizing the track.

channelsCount shall be the number of channels present in the soundField.

description of codec standard shall be a description of the standard. If the standard cannot be identified, then it shall be the string “Unknown”.

audioTrackKind should be one of the following value: “5.1Mix”, “METrack”, “Dialogue”. If another value is used, the values of the “typeNamespace” and “typeLink” attributes shall be changed.

channelLayout shall be a sequence of loudspeaker position abbreviations as specified in ISO/IEC 23091-3:2018, Table 2, second column, separated by a “ ” (space) character.

EXAMPLE 2 “L R C Lfe Ls Rs”.

channelLocation shall be the file number of a file in the soundPackage corresponding to the channelLayout separated by spaces.

EXAMPLE 3 “0 1 2 3 4 5”.

integratedLoudnessMethod should be “No”, or “R128”. If another value is used, the values of the “typeNamespace” and “typeLink” attributes shall be changed.

integratedLoudnessValue shall be loudness value according to the *integratedLoudnessMethod* indicated.

link to codec standard reference shall be a permanent link to the standard. If the standard cannot be identified, then it shall be the following URI: “urn:cen.eu:en17650:2022:ns:metadata#UnknownStandard”.

NOTE If no specific standard is used for the codec, it can be the same set of values as for the container.

11.3.1.5 Specific constraints for Timed Text Package

11.3.1.5.1 Constraints on the <coreMetadata> element

XPath is: “/ebuCore:ebuCoreMain/ebuCore:coreMetadata”.

The elements listed in 11.3.1.2.2 shall be present.

subPackageName shall be the name of the subpackage.

fileOriginTypeLink should be “urn:cen.eu:en17650:2022:ns:metadata#timedTextFileOrigin”.

If the *fileOriginTypeLink* is set to the recommended URI, *fileOriginContent* shall be one of the following values: “ManuallyCreated”, “AIGenerated”, “Unknown”.

The elements listed in Table 53 shall be present.

Table 53 — <coreMetadata> mandatory children for Timed Text Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	Value(s)
Language	language	1	@typeNamespace	x	<i>language definition namespace</i>
			@typeLink	x	<i>language definition link</i>
			#children	x	dc:language
	language/dc:language	1..n	#text	x	<i>Language description</i>
Format	format	1	@formatName	x	“timedTextPackageFormat”
			#children	x	see 11.3.1.5.2

language description, language definition namespace and language definition link shall follow the convention for language encoding. They should comply with the same standard recommendations as 11.1.2.

For each iteration language shall be the label of the language in the specified convention.

11.3.1.5.2 Constraints on the “timedTextPackageFormat” <format> element

XPath is:

“/ebuCore:ebuCoreMain/ebuCore:coreMetadata/ebuCore:format[@formatName='timedTextPackageFormat']”.

This element shall be present and shall be unique.

The elements listed in Table 54 shall be present.

Table 54 — “timedTextPackageFormat” <format> children for Timed Text Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Mime Type	mimeType	0..1	@typeLabel	x	<i>mime type</i>
Timed Text standard reference	technicalAttributeString	0..1	@typeLabel	x	“timedTextStandardReference”
			#text	x	<i>description of Timed Text standard</i>
Subtitling destination	technicalAttributeString	1	@typeLabel	x	“subtitleDestination”
			#text	x	<i>subtitle destination</i>
Character encoding	technicalAttributeString	0..1	@typeLabel	x	“characterEncoding”
			#text	x	<i>characterEncoding</i>
Using subpicture	technicalAttributeBoolean	0..1	@typeLabel	x	“useSubpictures”
			#text	x	<i>subpictures use</i>
Timed Text standard reference	technicalAttributeUri	0..1	@typeLabel	x	“timedTextStandardReference”
			#text	x	<i>link to Timed Text standard reference</i>

mime type shall be the MIME type of the file format.

EXAMPLE 1 for ST 428-7 subtitles, the MIME type is “application/xml”.

description of Timed Text standard shall be the name of the timed Text standard.

EXAMPLE 2 ST 428-7:2014 - SMPTE Standard - Digital Cinema Distribution Master — Subtitle.

subtitle destination should be one of the following values: “MainSubtitle”, “OpenCaption”, “ClosedCaption”.

For all subtitle text files, *characterEncoding* shall be “UTF-8”, according to 9.3.6.1.

subpicture use shall be “true” if at least one subpicture image is referenced in the Timed Text file, and “false” else.

link to Timed Text standard reference shall be a permanent link to the standard.

EXAMPLE 3 <https://doi.org/10.5594/SMPTE.ST428-7.2014>.

11.3.1.6 Specific constraints for Audiovisual Package

11.3.1.6.1 Constraints on the <coreMetadata> element

XPath is: “/ebuCore:ebuCoreMain/ebuCore:coreMetadata”.

The elements listed in 11.3.1.2.2 shall be present.

subPackageName shall be the name of the subpackage.

fileOriginTypeLink should be the *fileOriginTypeLinkDefaultValue*, as specified in 11.3.1.2.4.

If the *fileOriginTypeLink* is set to the recommended URI, *fileOriginContent* shall be one of the values included in *fileOriginContentDefaultValueList*, as specified in 11.3.1.2.4.

The elements listed in Table 55 shall be present.

Table 55 — <coreMetadata> mandatory children for Audiovisual Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	Value(s)
Package metadata	format	1	@formatName	x	“audiovisualPackageFormat”
			#children	x	see 11.3.1.6.2
Container metadata	format	1	@formatName	x	“containerFormat”
			#children	x	see 11.3.1.6.3
VideoStreams metadata	format	1..n	@formatName	x	“videoFormat”
			#children	x	see 11.3.1.6.5
AudioStreams metadata	format	0..n	@formatName	x	“audioFormat”
			#children	x	see 11.3.1.6.7

11.3.1.6.2 Constraints on the “audiovisualPackageFormat” <format> element

XPath is:

“/ebuCore:ebuCoreMain/ebuCore:coreMetadata/ebuCore:format[@formatName='audiovisualPackageFormat']”.

This element shall be present and shall be unique.

The elements listed in Table 56 shall be present.

Table 56 — “audiovisualPackageFormat” <format> children for Audiovisual Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Duration	duration	0..1	#children	x	normalPlayTime
	duration/normalPlayTime	1	#text	x	<i>normal playtime</i>
	duration	0..1	#children	x	timecode
	duration/timecode	1	#text	x	<i>duration timecode</i>

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Active part	start	0..1	#children	x	timecode
	start/timecode	1	#text	x	<i>start time code</i>
	end	0..1	#children	x	timecode
	end/timecode	1	#text	x	<i>end time code</i>

normal playtime shall be the duration of the content expressed as HH:MM:SS.S as specified by EBUCore.

duration timecode shall be the duration expressed in SMPTE ST 12-1:2008 timecode.

start time code shall be the start timecode expressed in SMPTE ST 12-1:2008 timecode.

end time code shall be the end time code expressed in SMPTE ST 12-1:2008 timecode.

11.3.1.6.3 Constraints on the “containerFormat” <format> element

XPath is:

“/ebuCore:ebuCoreMain/ebuCore:coreMetadata/ebuCore:format[@formatName='containerFormat']”.

This element shall be present and shall be unique.

The elements listed in Table 57 shall be present.

Table 57 — “containerFormat” <format> children for Audiovisual Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	Value(s)
Container format metadata	containerFormat	1	@containerFormatName	x	<i>formatName</i>
			@containerFormatDefinition	o	<i>formatVersionName</i>
			#children	x	See 11.3.1.6.4
MIME type	mimeType	0..1	@typeLabel	x	<i>mime type</i>

formatName shall be the name of the container format as specified in 11.3.1.2.3.

formatVersionName shall be a string denoting the version.

EXAMPLE “2.0”, “2018”.

mime type shall be the MIME type of the video file.

11.3.1.6.4 Constraints on the <containerFormat> element

XPath is:

“/ebuCore:ebuCoreMain/ebuCore:coreMetadata/ebuCore:format[@formatName='containerFormat']/ebuCore:containerFormat”.

The elements listed in Table 58 shall be present.

Table 58 — <containerFormat> children for Audiovisual Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Container standard reference	technicalAttributeString	1	@typeLabel	x	“containerStandardReference”
			#text	x	<i>description of container standard</i>
Container width	technicalAttributeUnsignedInteger	0..1	@typeLabel	x	“containerWidth”
			#text	x	<i>containerWidth</i>
Container height	technicalAttributeUnsignedInteger	0..1	@typeLabel	x	“containerHeight”
			#text	x	<i>containerHeight</i>
Container standard reference	technicalAttributeUri	1	@typeLabel	x	“containerStandardReference”
			#text	x	<i>link to container standard reference</i>

description of container standard shall be the name of the container standard. If the standard cannot be identified, then it shall be the string “Unknown”.

EXAMPLE 1 “ISO/IEC 14496-14:2020 - Information technology — Coding of audio-visual objects — Part 14: MP4 file format”.

containerWidth shall be the width of the video container. *containerHeight* shall be the height of the video container. This may not be the dimensions of the active area. If the subpackage encloses video, these attributes shall be present.

link to container standard reference shall be a permanent link to the standard. If the standard cannot be identified, then it shall be the following URI:

“urn:cen.eu:en17650:2022:ns:metadata#UnknownStandard”.

EXAMPLE 2 “https://www.iso.org/standard/79110.html”.

11.3.1.6.5 Constraints on the “videoFormat” <format> element

XPath is:

“/ebucore:ebuCoreMain/ebucore:coreMetadata/ebucore:format[@formatName='videoFormat']”.

The element listed in Table 59 shall be present.

Table 59 — “videoFormat” <format> children for Audiovisual Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	Value(s)
Video format metadata	videoFormat	0..1	#children	x	See 11.3.1.6.6

If the subpackage contains video, the videoFormat element shall be present.

11.3.1.6.6 Constraints on the <videoFormat> element

XPath is:

“/ebucore:ebuCoreMain/ebucore:coreMetadata/ebucore:format[@formatName='videoFormat']/ebucore:videoFormat”.

The elements listed in Table 60 shall be present.

Table 60 — <videoFormat> children for Audiovisual Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Region Delim X	regionDelimX	0..1	#text	x	<i>bottom left corner of the region</i>
Region Delim Y	regionDelimY		#text	x	<i>bottom left corner of the region</i>
Width	width		#text	x	<i>width</i>
Height	height		#text	x	<i>height</i>
Framerate	frameRate	1	@factorNumerator	o	<i>framerate numerator</i>
			@factorDenominator	o	<i>framerate denominator</i>
			#text	x	<i>nominal framerate</i>
Aspect ratio	aspectRatio	0..1	#children	x	<i>factorNumerator, factorDenominator</i>
	aspectratio/factorNumerator		#text	x	<i>ratioNumerator</i>
	aspectratio/factorDenominator		#text	x	<i>ratioDenominator</i>
Video Codec standard reference	technicalAttributeString	1	@typeLabel	x	<i>"videoCodecStandardReference"</i>
			#text	x	<i>description of video codec standard</i>
Colour standard reference	technicalAttributeString	1	@typeLabel	x	<i>"colourReference"</i>
			#text	x	<i>colour reference colour description</i>
Progressive Mode	technicalAttributeUnsignedByte	0..1	@typeLabel	x	<i>"progressiveMode"</i>
			#text	x	<i>{"0", "1", "2"}</i>
Chroma Subsampling	technicalAttributeUnsignedByte	0..1	@typeLabel	x	<i>"chromaSubsampling"</i>
			#text	x	<i>chroma subsampling value</i>
Number of colour components	technicalAttributeUnsignedInteger	1	@typeLabel	x	<i>"componentNumber"</i>
			#text	x	<i>Number of colour components</i>
Bitdepth of colour components	technicalAttributeUnsignedInteger	1	@typeLabel	x	<i>"componentBitdepth"</i>
			#text	x	<i>bitdepth</i>
Number of channels	technicalAttributeUnsignedInteger	1	@typeLabel	x	<i>"channelNumber"</i>
			#text	x	<i>channel number</i>
Index Colour Primaries	technicalAttributeUnsignedInteger	1	@typeLabel	x	<i>"colourPrimaries"</i>
			#text	x	<i>Index of colour primaries according ITU-T H.273</i>

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Index Transfer Characteristics	technicalAttributeUnsignedInteger	1	@typeLabel	x	"transferCharacteristics"
			#text	x	<i>Index of transfer characteristics according ITU-T H.273</i>
Index Matrix Coefficients	technicalAttributeUnsignedInteger	1	@typeLabel	x	"matrixCoefficients"
			#text	x	<i>Index of matrix coefficients according ITU-T H.273</i>
Video FullRange flag	technicalAttributeUnsignedInteger	1	@typeLabel	x	"fullRangeFlag"
			#text	x	<i>full range flag</i>
Luminance Min	technicalAttributeUnsignedInteger	0..1	@typeLabel	x	"luminanceMin"
			#text	x	<i>minimum Luminance of display</i>
Luminance Max	technicalAttributeUnsignedInteger	0..1	@typeLabel	x	"luminanceMax"
			#text	x	<i>maximum Luminance of display</i>
Value Range Min	technicalAttributeUnsignedInteger	0..1	@typeLabel	x	"valueRangeMin"
			#text	x	<i>minimum Value (black value)</i>
Value Range Max	technicalAttributeUnsignedInteger	0..1	@typeLabel	x	"valueRangeMax"
			#text	x	<i>maximum Value (white value)</i>
Colour standard reference	technicalAttributeUri	1	@typeLabel	x	"colourReference"
			#text	x	<i>colour reference colour link</i>
Video Codec standard reference	technicalAttributeUri	1	@typeLabel	x	"videoCodecStandardReference"
			#text	x	<i>link to video codec standard reference</i>

nominal framerate shall be filled with the nominal value of the framerate. In case of a fractional framerate, the correction shall be done in `factorNumerator` and `factorDenominator` attributes. If present, framerate numerator and framerate denominator shall be the rational numerator and denominator of the framerate.

EXAMPLE 1 for 24 fps, the *nominal framerate* is "24", and the *framerate numerator* and *framerate denominator* are not present, for 29.970 fps, the *nominal framerate* is "30", the *framerate numerator* is "1000", and *framerate denominator* is "1001".

description of video codec standard shall be a description of the video codec standard. If the standard cannot be identified, then it shall be the string "Unknown".

EXAMPLE 2 "ISO/IEC 14496-10:2009 Information technology — Coding of audio-visual objects — Part 10: Advanced Video Coding".

link to video codec standard reference shall be a permanent link to the standard. If the standard cannot be identified, then it shall be the following URI:

“urn:cen.eu:en17650:2022:ns:metadata#UnknownStandard”.

EXAMPLE 3 “https://www.iso.org/standard/52974.html”.

The other fields shall be filled in as specified in 0.

11.3.1.6.7 Constraints on the “audioFormat” <format> element

XPath is:

“/ebuCore:ebuCoreMain/ebuCore:coreMetadata/ebuCore:format[@formatName='audioFormat']”.

The element listed in Table 61 shall be present.

Table 61 — “audioFormat” <format> children for Audiovisual Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	Value(s)
Audio format metadata	audioFormat	1	#children	x	See 0

11.3.1.6.8 Constraints on the <audioFormat> element

XPath is:

“/ebuCore:ebuCoreMain/ebuCore:coreMetadata/ebuCore:format[@formatName='audioFormat']/ebuCore:audioFormat”.

The elements listed in Table 52 shall be present.

The fields shall be filled as specified in 11.3.1.4.6.

11.3.1.7 Specific constraints for Componentized Package

11.3.1.7.1 Constraints on the <coreMetadata> element

XPath is: “/ebuCore:ebuCoreMain/ebuCore:coreMetadata”.

The elements listed in 11.3.1.2.2 shall be present.

subPackageName shall be the name of the subpackage.

fileOriginTypeLink should be “urn:cen.eu:en17650:2022:ns:metadata#compFileOrigin”.

If the *fileOriginTypeLink* is set to the recommended URI, *fileOriginContent* shall be one of the following value: “FileBased”, “Unknown”.

The elements listed in Table 62 shall be present.

Table 62 — <coreMetadata> children for Componentized Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	Value(s)
Package metadata	format	1	@formatName	x	"componentizedPackageFormat"
			#children	x	see 11.3.1.7.2
Metadata for each CPL	format	1..n	@formatName	x	"cplFormat"
			#children	x	see 11.3.1.7.3

The <format> element which attribute `formatName` has value "cplFormat" shall be repeated as many times as there are CPLs present in the package.

11.3.1.7.2 Constraints on the "componentizedPackage" <format> element

XPath is:

"/ebucore:ebuCoreMain/ebucore:coreMetadata/ebucore:format[@formatName=componentizedPackageFormat]".

This element shall be present and shall be unique.

The elements listed in Table 63 shall be present.

Table 63 — "componentizedPackage" <format> children for Componentized Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Package type	technicalAttributeString	1	@typeLabel	x	"packageType"
			@typeNamespace	x	"en17650:2022"
			@typeLink	x	"urn:cen.eu:en17650:2022:ns:metadata#componentizedType"
			#text	x	<i>package type</i>
Package type	technicalAttributeString	0..1	@typeLabel	x	"packageSubtype"
			#text	x	<i>package subtype</i>
PKL Annotation text	technicalAttributeString	1	@typeLabel	x	"pklAnnotationText"
			#text	x	<i>PKL annotation text</i>
Referenced package	technicalAttributeString	0..n	@typeLabel	x	"referencedPackage"
			#text	x	<i>reference package</i>
Self-contained package	technicalAttributeBoolean	1	@typeLabel	x	"selfContainedPackage"
			#text	x	<i>self-contained package</i>

package type and package subtype shall be according to Table 64.

Table 64 — Package type and subtype values for Componentized Package

Enclosing package	type	subtype
DCP	"DCP"	"Interop"
		"MAP"
		"SMPTE"
IMF packages	"IMF"	"APP2"
		"APP2E"
		"APP4"
		"APP5"
		"RDD45"

PKL annotation text shall be the corresponding text of the AnnotationText element in the PKL file.

reference package shall be the UUID of the package containing the missing assets in the current package. Many packages may be indicated as this is the value of a repeatable element. If self-contained package is "false", then at least one package shall be referenced. If self-contained package is "true", no referencePackage element shall be present.

self-contained package shall be "true" if all assets referenced in all the CPLs are included in the package, and shall be "false" else.

11.3.1.7.3 Constraints on the "cplFormat" <format> element

XPath is:

"/ebuCore:ebuCoreMain/ebuCore:coreMetadata/ebuCore:format[@formatName='cplFormat']".

This element shall be repeatable.

The elements listed in Table 65 shall be present.

Table 65 — "cplFormat" <format> children for Componentized Package

Topic	EBUCore element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Annotation text	dc:format	0..1	#text	x	<i>annotation text</i>
VideoStreams metadata	videoFormat	1..n			
			#children	x	see Table 60
AudioStreams metadata	audioFormat	0..n			
			#children	x	see Table 52
Duration	duration	1	#children	x	editUnitNumber
	duration/editUnitNumber	1	#text	x	<i>number of edit units</i>
CPL Content title text	technicalAttributeString	1	@typeLabel	x	"cplContentTitleText"
			#text	x	<i>contentTitleText of the CPL</i>

annotation text shall be a free text describing the CPL.

number of edit units shall be duration of the CPL virtual track in number of edit units.

contentTitleText of the CPL shall be contentTitleText of the CPL.

11.3.2 Provenance metadata

The provenance metadata files shall conform to the XML standard and follow the PREMIS XML Schema 3.0.

12 Specification of the playlist

12.1 File content

Each playlist file shall contain the description of one playout.

The playlist shall describe either a timeline where the composition in time of Image, Sound and Timed Text Packages are referenced as elementary tracks, or partially assembled elements from Audiovisual Packages.

The CPL format is derived from the IMF standard. The timeline is divided in segments, for instance film reels. Multiple sequences may be present in a segment, each of them defining a virtual track. Each sequence can be composed, sequentially, by one or more resources with entry points and duration.

EXAMPLE As this standard playlist schema provides ImageASequence and ImageBSequence sequenceTypes, so a segment could contain an ImageASequence containing a link to the imagePackage resource corresponding to image A, and an ImageBSequence containing a link to the imagePackage resource corresponding to image B.

12.2 File conformance

Each playlist file shall conform to the XML standard and follow the IMF Composition Playlist format as specified in SMPTE ST 2067-3:2020 and with the XML schema specified in Annex B.

12.3 Constraints on the Playlist XML file

12.3.1 EssenceDescriptor elements

The EssenceDescriptorList element shall not be present.

12.3.2 Resource elements

The XML schema in Annex B has a specific CppTrackFileResourceType that shall be used as the xml schema type for the Resource element within the ResourceList element.

This Resource element shall contain a CppPackingListId element, and the child text of this element shall be the *subpackageId* in the packingList, as specified in 8.4.4.6, which maps the UUID specified in 6.4.3.

EXAMPLE "imagePackage_6e179467-f1c6-480f-b8f0-2c8c993cce36".

12.3.3 Sequence elements

In the SequenceList, for each Sequence, one of the following virtual tracks elements should be used:

- AudioSequence;
- SubtitleSequence;
- CaptionSequence;

- ImageSequence;
- ImageASequence;
- ImageBSequence.

12.4 File revision

The unique identifier (UUID) within the playlist file shall be changed on each update of the contained list and values.

13 Checker reports

13.1 General description (informative)

Checker reports files result of a test procedure after the package was created. The test procedure can execute completeness checks, tests on the existence of mandatory elements, lists of non-standardised elements, lists of used standards and so on.

13.2 Constraints on the XML Checker Reports List files

13.2.1 General description (informative)

A Checker Reports List file enumerates all checker report files with additional metadata on the product used to generate the report.

13.2.2 General constraints

The Checker Reports List file shall be an XML file and shall comply with the XML Schema in Annex C.

13.2.3 Constraints on the checkerDescription element

The elements listed in Table 66 are specified in the XML schema and repeated here for convenience.

Table 66 — checkerDescription children Checker Reports List

Topic	XML element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Checker description	vendor	1	#text	x	<i>vendor name</i>
	productName	1	#text	x	<i>product name</i>
	productMajorVersion	1	#text	x	<i>product major version</i>
	productMinorVersion	1	#text	x	<i>product minor version</i>

vendor name shall be the name of the checker vendor name.

product name shall be the name of the checker product.

product major version shall be the product major version number.

product minor version shall be the production minor version number.

13.2.4 Constraints on the checkerProcessDescription element

The elements listed in Table 67 are specified in the XML schema and repeated here for convenience.

Table 67 — checkerProcessDescription children for Checker Reports List

Topic	XML element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Checker process description	processID	1	#text	x	<i>processID</i>
	creationTimestamp	1	#text	x	<i>creation timestamp</i>
	checkedPreservationPackage	1	#text	x	<i>checked preservation package</i>
	checkResult	1	#text	x	<i>check result</i>
	resultString	0..1	#text	x	<i>check result description</i>

processID shall be a new UUID for each new check process.

creation timestamp shall be the ISO 8601 date and time representation of report creation.

EXAMPLE <creationTimestamp>2020-11-23T12:00:00</creationTimestamp>.

checked preservation package shall be the UUID of the CPP package, as specified in 6.9.

check result shall be “true” if the check process succeed without error, “false” otherwise.

check result description shall be an overall description of the result.

13.2.5 Constraints on the checkerFileDescription element

The elements listed in Table 68 are specified in the XML schema and repeated here convenience.

Table 68 — checkerFileDescription children for Checker Reports Lists

Topic	XML element	Element Arity	Attribute or other subelement	Subelement mandatory	value
Checker file description	location	1	#text	x	<i>location</i>
	mimetype	1	#text	x	<i>mime type</i>
	checksum	0..n	#children	x	<i>checksumValue, checksumAlgorithm</i>
	checksum/checksumValue	1	#text	x	<i>checksum</i>
	checksum/checksumAlgorithm	1	#text	x	<i>checksum algorithm</i>
	fileDescription	0..1	#text	x	<i>file description</i>

location shall be the location of the checker report file.

mime type shall be the MIME type of the file.

checksum shall be the hash value of the file computed with the checksum algorithm.

file description shall be a user description of the file.

Annex A (normative)

XML Schema for the Packing List

The following schema defines attributes to be used into the preservationPackingList.xml file, or in the subpackage packingList.xml files. These attributes are meant to be used as additional attributes into the <mets:metsHdr> element.

```
<?xml version="1.0" encoding="UTF-8"?>
<schema
  xmlns="http://www.w3.org/2001/XMLSchema"
  targetNamespace="urn:cen.eu:en17650:2022:ns:packingList"
  xmlns:tns="urn:cen.eu:en17650:2022:ns:packingList"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  elementFormDefault="qualified">

  <xs:simpleType name="packingFileType" final="restriction">
    <xs:restriction base="xs:string">
      <xs:enumeration value="preservationPackingList" />
      <xs:enumeration value="subpackagePackingList" />
    </xs:restriction>
  </xs:simpleType>
  <attribute name="packingList" type="tns:packingFileType"></attribute>

  <xs:simpleType name="packageProfileType" final="restriction">
    <xs:restriction base="xs:string">
      <xs:enumeration value="interimPreservationProfile" />
      <xs:enumeration value="longTermPreservationProfile" />
    </xs:restriction>
  </xs:simpleType>
  <attribute name="packageProfile" type="tns:packageProfileType"></attribute>

  <attribute name="packageFormatVersion" type="anyURI"></attribute>

</schema>
```

Annex B (normative)

XML Schema for the Playlist

The following schema extends the IMF CPL Schema as defined in SMPTE ST 2067-3:2016 Interoperable Master Format — Composition Playlist. The various sequences may be used to describe one particular sequence.

The *CppTrackFileResourceType* should be used to reference one subpackage present in the current Cinema Preservation Package.

```
<?xml version="1.0" encoding="UTF-8"?>
<schema xmlns="http://www.w3.org/2001/XMLSchema"
  targetNamespace="urn:cen.eu:en17650:2022:ns:playlist"
  xmlns:tns="urn:cen.eu:en17650:2022:ns:playlist"
  xmlns:cpl="https://www.smp-te-ra.org/schemas/2067-3/2016"
  xmlns:dcml="https://www.smp-te-ra.org/schemas/433/2008/dcmlTypes/"
  xmlns:ds="http://www.w3.org/2000/09/xmldsig#"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  elementFormDefault="qualified" attributeFormDefault="unqualified">
  <xs:import namespace="https://www.smp-te-ra.org/schemas/2067-3/2016" />
  <xs:import namespace="https://www.smp-te-ra.org/schemas/433/2008/dcmlTypes/" />

  <xs:element name="ImageSequence" type="cpl:SequenceType" />
  <xs:element name="ImageASequence" type="cpl:SequenceType" />
  <xs:element name="ImageBSequence" type="cpl:SequenceType" />
  <xs:element name="AudioSequence" type="cpl:SequenceType" />
  <xs:element name="SubtitleSequence" type="cpl:SequenceType" />
  <xs:element name="CaptionSequence" type="cpl:SequenceType" />

  <xs:simpleType name="CppIdType">
    <xs:restriction base="xs:string">
      <xs:pattern value="[a-zA-Z]+_[0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12}(_[0-9]{3,8})?" />
    </xs:restriction>
  </xs:simpleType>

  <complexType name="CppTrackFileResourceType">
    <xs:complexContent>
      <xs:extension base="cpl:BaseResourceType">
        <xs:sequence>
          <xs:element name="CppPackingListId" type="tns:CppIdType" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </complexType>
</schema>
```

Annex C (normative)

XML Schema for the Checker Reports List

```

<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
  xmlns="urn:cen.eu:en17650:2022:ns:checkerReportsList"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  targetNamespace="urn:cen.eu:en17650:2022:ns:checkerReportsList">
  <xs:element name="checkerResultRegistry" type="CheckerResultRegistryType" />
  <xs:complexType name="CheckerResultRegistryType">
    <xs:sequence>
      <xs:element name="checkerResult" type="checkerResultType" minOccurs="0"
maxOccurs="unbounded" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="checkerResultType">
    <xs:sequence>
      <xs:element name="checkerDescription" type="checkerDescriptionType" />
      <xs:element name="checkerProcessDescription"
type="checkerProcessDescriptionType" />
      <xs:element name="checkerFileDescription" type="checkerFileDescriptionType"
minOccurs="1" maxOccurs="unbounded" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="checkerDescriptionType">
    <xs:sequence>
      <xs:element name="vendor" type="xs:string" />
      <xs:element name="productName" type="xs:string" />
      <xs:element name="productMajorVersion" type="xs:string" />
      <xs:element name="productMinorVersion" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="checkerProcessDescriptionType">
    <xs:sequence>
      <!-- Unique ID for the checking process, this ID shall be used as subdirectory
name inside the Root Checker Reports folder -->
      <xs:element name="processID" type="UUID" />
      <xs:element name="creationTimestamp" type="xs:dateTime" />
      <xs:element name="checkedPreservationPackage" type="UUID" />
      <xs:element name="checkResult" type="xs:boolean" />
      <xs:element name="resultString" type="xs:string" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="checkerFileDescriptionType">
    <xs:sequence>
      <xs:element name="location" type="xs:string" />
      <xs:element name="mimetype" type="xs:string" />
      <xs:element name="checksum" type="checksumType" minOccurs="0" />
      <xs:element name="fileDescription" type="xs:string" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="checksumType">
    <xs:sequence>
      <xs:element name="checksumValue" type="xs:string" />
      <xs:element name="checksumAlgorithm" type="xs:string" />
    </xs:sequence>
  </xs:complexType>

```

```
</xs:complexType>
<xs:simpleType name="UUID">
  <xs:restriction base="xs:string">
    <xs:pattern
      value="([0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12})|(\{[0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12}\})" />
    </xs:restriction>
  </xs:simpleType>
</xs:schema>
```

Annex D (normative)

Tables from ISO 21122-3:2019

D.1 Interlaced modes

Table D.1 shows the interlaced mode.

Table D.1 — Format of the contents of frat Interlace_Mode

Field value	Meaning
0	Progressive frame (frame contains one full-height picture)
1	Interlaced frame (first picture is top field)
2	Interlaced frame (second picture is top field)
3	Reserved

D.2 Sampling structure

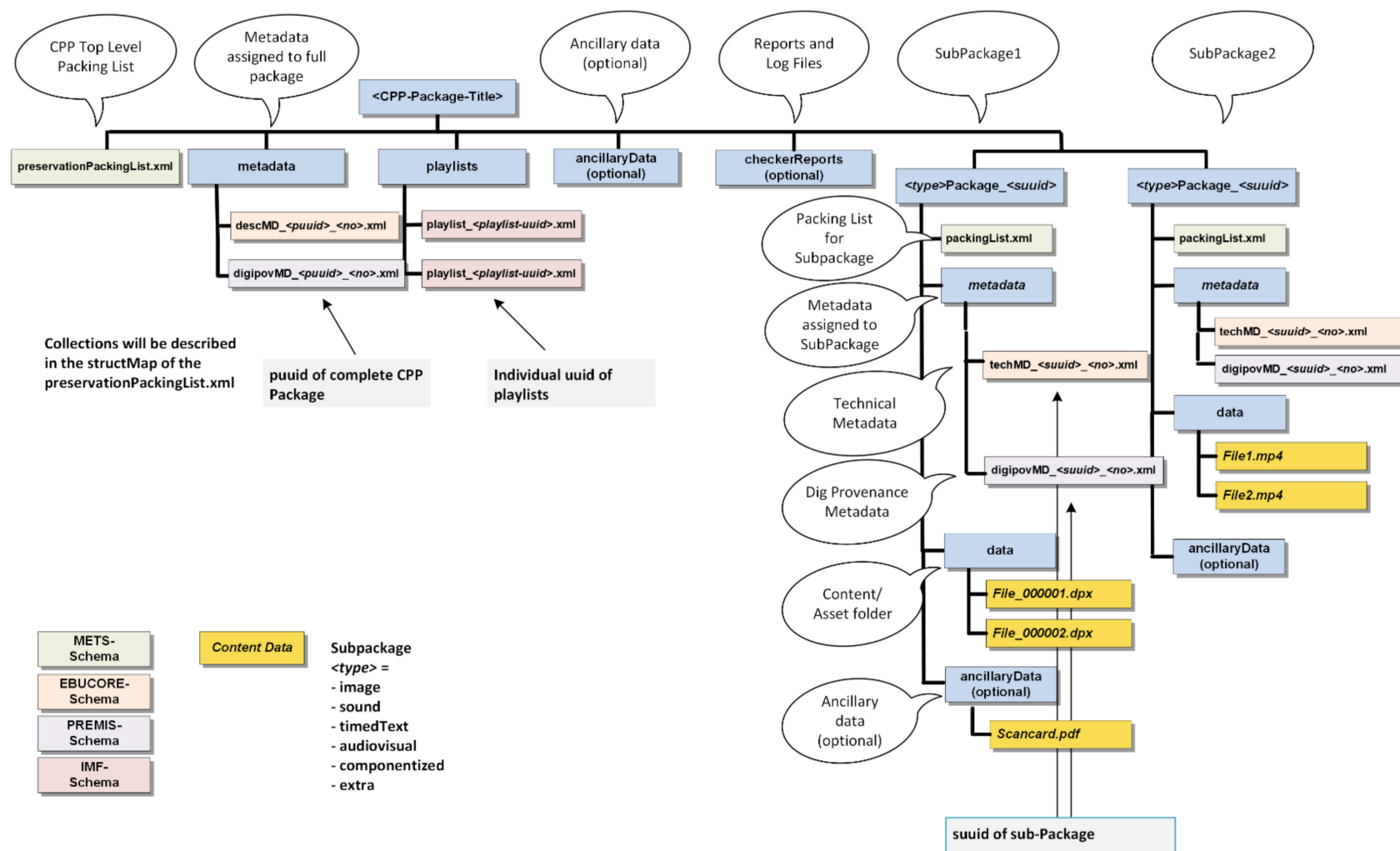
Table D.2 shows the sampling structure (see ISO 21122-3).

Table D.2 — Index values for subsampling formats

Field value	Meaning	N_c
0	4:2:2 (YCbCr)	3
1	4:4:4 (YCbCr)	3
2	4:4:4 (RGB)	3
3	4:2:0 (YCbCr)	3
4	4:2:2:4 (YCbCrAux)	4
5	4:4:4:4 (YCbCrAux)	4
6	4:4:4:4 (RGBAux)	4
7-15	Reserved	Reserved

Annex E
(informative)

Synthetic package view



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